ACADEMIC REGULATIONS & SYLLABUS

(Choice Based Credit System)

Faculty of Technology & Engineering (FTE)
Chandubhai S. Patel Institute of Technology (CSPIT)

Master of Technology Programme
Advanced Manufacturing Technology (AMT)
Mechanical Engineering

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology and Engineering

ACADEMIC REGULATIONS

Master of Technology (Advanced Manufacturing Technology) Programme

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, Off. Nadiad-Petlad Highway, Changa 388 421, Anand, Gujarat, INDIA Telephone: 02697-265011 & 02697-265021, Email: info@charusat.ac.in
www.charusat.ac.in

Academic Year 2024-25

CHARUSAT

FACULTY OF TECHNOLOGY AND ENGINEERING
ACADEMIC REGULATIONS
Master of Technology Programmes
Choice Based Credit System

Academic regulations recommendations are provided to ensure uniform system of education, programmes duration, eligibility criteria for admission, course credits distribution, teaching and examination pedagogy, detailed syllabus with reference material.

1) System of Education

   The Charotar University of Science and Technology (CHARUSAT) shall follow Choice based Credit System (CBCS) with Semester pattern at Undergraduate and Master levels. Each semester will be at least of 90 working days. Apart from the programme core courses, provision for choosing University level electives and Programme electives are available under the CBCS.

2) Duration of Programme

(i)	Postgraduate programme	(M.Tech)
	Minimum	4 semesters (2 academic years)
	Maximum	6 semesters (3 academic years)

3) Eligibility for admissions

   As enacted by Government of Gujarat/AICTE/UGC from time to time.

4) Mode of admissions

   As enacted by Government of Gujarat from time to time.

5) Programme Structure and Credits

   As per annexure – 1 attached

6) Attendance

   6.1. Students are expected to maintain 100% attendance in all courses. However, students may involuntarily have to miss classes due to illness or some family emergency; students are permitted to maintain a minimum attendance of 75% with producing proof or reason for the absence. In case of medical exigencies, the student/parent should inform the principal immediately through call or by email. Within a week, starting from the day of absence, the proof of medical exigency must be submitted to the Principal's office.

6.2. Unauthorized absence will be considered as part of the discretionary 25% for fulfilling the minimum 75% attendance requirement for appearing in the examination.

6.3. Students nominated/sponsored by the University to represent in various forums like seminars/conferences/workshops/competitions or taking part in co-curricular/extra-curricular events will be given attendance credit provided the student applies in writing for such a leave in advance and obtains sanction from the Principal of his/her Institute for academic related requests.

7) Course Evaluation

7.1. The performance of every student in each course will be evaluated as follows:

7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment. The respective department /institute will conduct the continuous assessment. The course faculty members shall share the pedagogy related to the continuous evaluation with the students.

7.1.2 Final end-semester examination shall be conducted by the University through written paper, practical test, oral test, presentation by the student or a combination of these.

7.1.3 The weightages of continuous assessment and end-semester university examination in overall assessment shall depend on individual course as approved by Academic Council through Faculty Board and Board of Studies.

7.1.4 The performance of candidate in continuous assessment and in end-semester examination together (if applicable) shall be considered for deciding the final grade in a course.

7.1.5 In order to earn the credit in a course a student has to obtain grade other than FF.

7.2. Performance in continuous assessment and end-semester University Examination

7.2.1 Minimum performance with respect to continuous assessment as well as end-semester university examination will be an important consideration for passing a course.

7.2.2 If a candidate fails to obtain minimum required overall percentage of marks (36%), student has to repeat the examination till the minimum required overall percentage obtained.

8) Grade Point System

1. The total of the internal evaluation marks and end semester examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:
2. Proposed Grading Scheme to awarding letter grade and grade point as per NEP 2020

Letter Grade	Grade Point	Grading Scheme for Mark (In %)
O (Outstanding)	10	96.0-100
A+ (Excellent)	9	86.0-95.9
A (Very Good)	8	76.0-85.9
B+ (Good)	7	66.0-75.9
B (Above Average)	6	56.0- 65.9
C (Average)	5	46.0 – 55.9
P (Pass)	4	36.0 – 45.9
F (Fail)	0	Below 36.0
Ab (Absent)	0	Absent

- The minimum passing marks for each pattern of evaluation are 36%

3. The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his/her performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

(i) SGPA = Σ Ci Gi / Σ Ci where Ci is the number of credits of course i

Gi is the Grade Point for the course i
and i = 1 to n, n = number of courses in the semester

(ii) CGPA = Σ Ci Gi / Σ Ci where Ci is the number of credits of course i

Gi is the Grade Point for the course i
and i = 1 to n, n = number of courses of all semesters up to which CGPA is computed.

9) Award of Class

- ❖ The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Award of Class	**CGPA Range**
First Class with Distinction	CGPA ≥ 7.0 & ≤ 10.0
First class	CGPA ≥ 6.0 & < 7.0
Second Class	CGPA ≥ 5.0 & < 6.0
Pass Class	CGPA < 5.0

10) Detention Criteria

- ❖ A student will be promoted to next year only if he/she has cleared all the courses of the year he/she is studying in.

  Link: https://charusat.ac.in/ => Student's Corner => Detention Rules

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
(CHARUSAT)

FACULTY OF TECHNOLOGY & ENGINEERING (FTE)

CHOICE BASED CREDIT SYSTEM

FOR

MASTER OF TECHNOLOGY

Choice Based Credit System

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist Foundation courses, Core courses, Elective courses, Non-credit (audit) courses and SWAYAM MOOCs.

1.1 Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

1.2 Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme. Environmental science will be a compulsory University core for all Undergraduate Programmes.

B. Programme Core courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

1.3 Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialised or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties / programmes. As a general guideline, Programme should incorporate 2 University Electives of 2 credits each (total 4 credits).

B. Programme Elective Courses(PE):

The programme specific pool of elective courses offered by respective programme.

1.4 Non Credit Course (NC) - AUDIT Course

A 'Non Credit Course' is a course where students will receive Participation or Course Completion certificate. This will be reflected in Student's Grade Sheet but the grade of the course will not be considered to calculate SGPA and CGPA. Attendance and Course Assessment is compulsory for Non Credit Courses.

1.5 Credit Transfer through SWAYAM MOOCs

CHARUSAT provides credit transfer as per UGC guidelines to all the students from SWAYAM against elective courses. The credit transfer is offered in two modes: (a) Partial credit transfer (b) Full credit transfer.

1.6 Medium of Instruction

The Medium of Instruction will be English.

In consonance with the National Education Policy (NEP) 2020 and the guidelines of the University Grants Commission (UGC), Charotar University of Science and Technology (CHARUSAT) implements the Multiple Exit scheme in their Master of Technology programme. Table 1 shows the exit qualifications along with credit requirements.

Table 1 Exit Qualifications along with Credit Requirements

ACADEMIC LEVEL	EXIT QUALIFICATION AND CREDITS REQUIRED	NATIONAL CREDIT LEVEL (NCrF)
1st year of PG Degree	Post Graduate Diploma will be awarded after completion of 1st year of 2-year PG program *Minimum of 40 credits for individuals who have completed a bachelor's programme*	6.5

Exit after First Year:

Award: PG Diploma in XXXX

The student shall be awarded with "PG Diploma in XXXX", with redemption of credits from ABC. Total credits redemption shall be 1st year credits.

Department of Mechanical Engineering – Chandubhai S Patel Institute of Technology

Vision

To be recognized as a center for outstanding education and research in field of mechanical engineering.

Mission

To produce well qualified engineers, who are innovative, contributors to their profession by catering to diverse societal needs.

Program Educational Objectives:

1. To prepare graduates having sound technical knowledge and core competency in the field of manufacturing technology.
2. To prepare graduates having effective communication, leadership, problem-solving, and decision making skills, supported by analytical, research and design skills to expand their knowledge; contributing to their overall personality, career development and life-long learning.
3. To prepare graduates exhibiting professional and ethical responsibilities towards their peers, employers, and society.

Program Outcomes:

PO1: An ability to independently carry out research /investigation and development work to solve practical problems.
PO2: An ability to write and present a substantial technical report/document.
PO3: Students would be able to work individually and in the team for betterment of the society.

Program Specific Outcomes:

PSO1: Students will be able to demonstrate leadership, effective communication, research potential and decision-making skills to analyse and solve manufacturing engineering problems.

Annexure-1

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
PROPOSED TEACHING AND EXAMINATION SCHEME FOR M TECH AMT
CHOICE BASED CREDIT SYSTEM SCHEME (2024-25)

Sem.	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Theory	Practical	Total	Credit	Theory		Practical		Total
							Internal	External	Internal	External	
Sem. 1	MEUC551	Casting & Welding Processes	4	2	6	5	50	50	25	25	150
	MEUC552	Unconventional Machining Processes	3	2	5	4	50	50	25	25	150
	MEUC553	Computer Aided Engineering	3	2	5	4	50	50	25	25	150
	MEUE5XX	Programme Elective-I (PE-1)	3	2	5	4	50	50	25	25	150
	MEUE5XX	Programme Elective-II (PE-2)	3	2	5	4	50	50	25	25	150
	HSUS501/ HSUS502	Academic Speaking And Presentation Skills/Foreign Languages (FRENCH)	2		2	2	00	00	25	25	50
					28	**23**					**800**
Sem. 2	MEUC554	Machining & Forming Processes	4	2	6	5	50	50	25	25	150
	MEUC555	CNC Technology & Programming	3	0	3	3	50	50	0	0	100
	MEUC556	Industrial Automation & Robotics	3	0	3	3	50	50	0	0	100
	MEUS551	Automation Laboratory	0	2	2	1	0	0	25	25	50
	MEUA551	Seminar	0	2	2	1	0	0	25	25	50
	MEUE5XX	Programme Elective-III (PE-3)	3	2	5	4	50	50	25	25	150
	MEUE5XX	Programme Elective-IV (PE-4)	3	2	5	4	50	50	25	25	150
	HSUA501	Academic Writing	0	2	2	2	00	00	25	25	50
		University Elective-II	2		2	2	00	00	25	25	50
					30	**25**					**950**

PE: Programme Elective   UC: University Core

Semester	Course Code	Course Title	Credit	Examination Scheme					
				Internal		External			Total
				Progress Report	Progress Seminar	Report	Seminar	Viva	
Sem. 3	MEUP601	Project Preliminaries	4	25	25	--	25	25	100
	MEUP602	Project Phase – I	18	100	150	50	100	100	500
			22						**600**
Sem. 4	MEUP603	Project Phase - II	18	100	150	50	100	100	500
			18						**500**
		Grand Total	**88**						**1100**

Programme Electives (PEC):

Sr. No.	Course Nature	Course Code	Course Title
1	Programme Elective-I (PE-1)	MEUE551	Advanced Materials
2		MEUE552	Experimental Techniques
3		MEUE553	Industry 4.0 & Industry IOT
4	Programme Elective-II (PE-2)	MEUE554	Manufacturing Planning & Control
5		MEUE555	Advanced Metrology
6	Programme Elective-III (PE-3)	MEUE556	Surface Engineering
7		MEME551	Computational Fluid Dynamics of Incompressible flows
8	Programme Elective-IV (PE-4)	MEUE557	Advanced Engineering Optimization Techniques
9		MEUE558	Artificial Intelligence
10		MEME552	Mechanics of Fiber Reinforced Polymer Composite Structures

M. Tech. (AMT) Programme

SYLLABI

(Semester – 1)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUC551: CASTING AND WELDING PROCESSES
M TECH 1st SEM ADVANCED MANUFACTURING TECHNOLOGY

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

A Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to casting and welding processes	02
2.	Solidification phenomena	12
3.	Defects in casting	05
4.	Tooling and Methoding design and analysis in casting	15
5.	Advances in Foundry Practices	07
6.	Physics of welding	10
7.	Weld defects and failures	05
8.	Advances in materials processing	04

Total hours (Theory): 60

Total hours (Practical): 30

Total hours: 90

B Detailed Syllabus:

1 Introduction to casting and welding processes 02 Hours 03%

1.1 Working principle, process capabilities of various casting processes.

1.2 Applications.

2	**Solidification phenomena**	**12 Hours**	**20%**
2.1	Liquid solid transformation, nucleation, crystal growth.		
2.2	Solidification of castings, rapid solidification, progressive and directional solidification, microstructure and cooling stress, influence of grain size and morphology.		
2.3	Analytical solutions of solidification heat transfer.		
3	**Defects in casting**	**05 Hours**	**10%**
3.1	Defects arising with various casting processes, their identification and preventing methods.		
4	**Tooling and Methoding design and analysis in casting**	**15 Hours**	**25%**
4.1	Part orientation and parting, Pattern allowances, Pattern design		
4.2	Cored hole features, Core print design and analysis		
4.3	Shape complexity, Multi cavity layout		
4.4	Feeder location and shape, feeder and neck design		
4.5	Applications of feed-aids to achieve directional solidification		
4.6	Gating system and types, Gating channel layout,		
4.7	Ideal and optimal filling time, Gating element design		
5	**Advances in Foundry Practices**	**07 Hours**	**11%**
5.1	Casting design and analysis using CAD software		
5.2	Design for castability		
5.3	SMART Foundry		
5.4	Rapid tooling in casting		
6	**Physics of welding**	**10 Hours**	**15%**
6.1	Characteristics of arc and mode of metal transfer, welding fluxes and coatings arc length and arc efficiency, melting efficiency in welding.		
6.2	Electrode codes and their critical evaluation.		
7	**Weld defects and failures**	**05 Hours**	**10%**
7.1	Fusion weld defects, like porosity, inclusion, weld cracking, failure, weld metal solidification cracks, hot cracks, heat affected zone, off shore structure, base metal defects.		

7.2 Stresses in heat affected zone, residual stresses, factors affecting residual stresses and stress corrosion cracking, relief of residual stresses, detection of weld defects.

7.3 Weld distortion, design consideration to minimize distortion.

8	**Advances in materials processing**	**04 Hours**	**6%**
8.1	Solid state processing of materials		
8.2	Liquid state processing of materials		

C Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.
- Industrial visits will be organized for students to explore industrial facilities. Students are required to prepare a report on industrial visit and submit as a part of the assignment.

D Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Design elements of tooling systems like pattern, mold and core.

CO2: Compute the solidification time in sand and metal die casting.

CO3: Evaluate and design casting feeding and gating system.

CO4: Analyze welding processes.

CO5: Identify the defects occur in casting and welding processes and provide the remedial suggestions for elimination.

CO6: Explore of the use of CAD software and IOT for casting and welding processes.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	3	1	2	2
CO2	1	-	-	1
CO3	2	2	3	2
CO4	2	-	-	2
CO5	2	2	-	3
CO6	1	-	1	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

E Recommended Study Material:

Text Books/ Reference Book:

1. P. C. Mukherjee, "Fundamentals of Metal Casting Technology", Oxford & IBH Pub. Co.
2. B. Ravi, "Metal Casting: Computer-aided Casting Design and Analysis"; PHI.
3. D.R. Poirier and E.J. Poirier, "Heat transfer Fundamentals for Metal Casting", Wiley.
4. S. Kalpakjian and S. Schmid, "Manufacturing Processes for Engineering Materials" Prentice Hall.
5. ASM Handbook Volume 15: Casting.
6. R.S.Parmar, "Welding Processes and Technology", Khanna Publishers.
7. R. Bittle, "Welding Technology", TMH.
8. William A. Bowditch, Kevin E. Bowditch, Mark A. Bowditch, "Welding Technology Fundamentals", Goodheart-Willcox Co.
9. R. W. Messler, "Principles of Welding", John Wiley & Sons.
10. O. Grong, "Metallurgical Modelling of Welding", IOM Publication.
11. P.T. Houldcorft, "Welding Process Technology", Cambridge University Press Metallurgy of Welding, Brazing and Soldering by J.F. Lancaster; George Allen & Unwin, London.
12. ASM Handbook Volume 6: Welding, Brazing, and Soldering.

Reading Materials, web materials with full citations:

1. http://nptel.iitm.ac.in/courses/Webcoursecontents/IIT%20Kharagpur/Manuf%20Proc%20II/New_index1.html
2. http://www.sciencedirect.com/science/journal
3. http://www.ieindia.org/publish/pr/pr.htm
4. http://www.ias.ac.in/sadhana

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUC552 : UNCONVENTIONAL MACHINING PROCESSES
M TECH 1st SEM ADVANCED MANUFACTURING TECHNOLOGY

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction	02
2.	Mechanical energy based processes	13
3.	Thermal & electrothermal energy based processes	14
4.	Chemical & electrochemical energy based processes	06
5.	Hybrid processes	10

Total hours (Theory): 45
Total hours (Practical): 30
Total hours: 75

B Detailed Syllabus:

1 Introduction **02 Hours** **5%**

1.1 Need for non-traditional machining methods, classification of modern machining process, consideration on process selection, application.

2 Mechanical energy based processes **13 Hours** **30%**

2.1 AJM- USM- WJM- basic principles, equipments used, process variables, process characteristics mechanics of metal removal, application and limitations, recent development .

3 Thermal & electrothermal energy based processes 14 Hours 30%

3.1 EDM- working principal equipments, process parameters, surface finish and MRR, electrode/tool, power and control circuits, tool wear, application and limitations, recent development.

3.2 LBM- LBD- IBM- EBM- PAM- principles, equipments, comparison of thermal and non thermal process, thermal features, process characteristics, types beam/laser, process variables, MRR, control techniques, application and limitations, recent development.

4 Chemical & electrochemical energy based processes 06 Hours 15%

4.1 CHM- ECM-ECG- principles, theory, equipments, process parameters, process characteristics, surface finish and MRR, tool, maskants, etchant, recent development.

5 Hybrid processes 10 Hours 20%

5.1 AWJM-AFF-MAF-MRF-FIBM-ElectroSream Drilling, Shaped Tube Electrolytic Machining-introduction, working principles, process performance, application.

C Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behaviour will be observed strictly.
- Industrial visits will be organized for students to explore industrial facilities. Students are required to prepare a report on industrial visit and submit as a part of the assignment.

D Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Summarize the needs and classification of unconventional machining process.

CO2: Explain the various input and output parameters that influence in the performance.

CO3: Explain the working principle of unconventional machining process.

CO4: Compare the merits, demerits and applications of unconventional machining process.

CO5: Perform the electric discharge machining.

CO6: Calculate the material removal rate for unconventional machining process.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	1	-	-
CO2	2	2	-	1
CO3	2	-	-	1
CO4	2	2	1	2
CO5	3	-	-	2
CO6	2	-	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

E Recommended Study Material:

Text Books/ Reference Book:

1. Jain V K, Advanced Machining Processes, Allied Publishers, Mumbai
2. Benedict. G.F. "Nontraditional Manufacturing Processes" Marcel Dekker Inc., New York
3. Pandey P.C. and Shan H.S. "Modern Machining Processes" Tata McGraw-Hill, New Delhi
4. Mc Geough, "Advanced Methods of Machining" Chapman and Hall, London.
5. P K Mishra, Unconventional Machining Process
6. Paul De Garmo, J.T.Black, and Ronald.A.Kohser, "Material and Processes in Manufacturing" Prentice Hall of India Pvt. Ltd., New Delhi
7. Adityan, Modern Machining Process

Reading Materials, web materials with full citations:

1. http://nptel.iitm.ac.in/
2. Mini Tab 15.0, Design Expert 7.0, Statgraphics
3. Sadhna: http://www.ias.ac.in/sadhana/
4. Mechanical Engg. (Inst. of Engineers) http://www.ieindia.org/publish/mc/mc.htm
   Production Engg. (Inst. of Engineers) http://www.ieindia.org/publish/pr/pr.htm
5. IEEE transactions on Manufacturing Technology
   http://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=8218
6. IET Manufacturing Engineer http://ieeexplore.ieee.org/servlet/opac?punumber=2189
7. Journal of Material Processing Technology (Elsevier Publication)

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MEUC553: COMPUTER AIDED ENGINEERING
M TECH 1st SEM ADVANCED MANUFACTURING TECHNOLOGY

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Fundamentals of CAE	03
2.	Fundamentals of computer graphics	10
3.	Geometric modeling	08
4.	Modeling of manufacturing processes	04
5.	Numerical solution techniques	16
6.	Reverse engineering and rapid manufacturing	04

Total hours (Theory): 45

Total hours (Practical): 30

Total hours: 75

B Detailed Syllabus:

1 Fundamentals of CAE **03 Hours** **07%**

1.1 Introduction to CAE and integration of design process.

1.2 Application and benefits of CAE.

1.3 Standards in CAD, graphics and computing standards.

1.4 Data exchange standards, design database.

2 Fundamentals of computer graphics **10 Hours** **22%**

2.1 Introduction to computer graphics.

2.2 Scan conversions and algorithm for generation of line and circle.

2.3 2D and 3D transformations.

3	**Geometric modeling**	**08 Hours**	**18%**
3.1	Representation of curves and surfaces.		
3.2	Geometric modeling techniques.		
3.3	Wireframe modeling, surface modeling and solid modeling.		
3.4	Feature based parametric and variation modeling.		
4	**Modeling of manufacturing processes**	**04 Hours**	**09%**
4.1	Mathematical modeling of manufacturing process.		
4.2	Physics involved and their governing Equations.		
4.3	Assumptions and approximations in the models.		
5	**Numerical solution techniques**	**16 Hours**	**35%**
5.1	Finite element method		
5.2	Finite difference and volume method		
6	**Reverse engineering and rapid manufacturing**	**04 Hours**	**09%**
6.1	Reverse engineering and its applications.		
6.2	Point cloud data acquisition and surface modelling		
6.3	Fundamentals and applications of rapid prototyping.		
6.4	Rapid manufacturing process chain.		

C Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Explain concepts, techniques and applications of computer aided engineering.

CO2: Develop program for geometric transformation and computer graphics.

CO3: Apply different techniques to create 2D & 3D geometric models.

CO4: Describe different methods of rapid prototyping and reverse engineering.

CO5: Implement numerical methods to simulate manufacturing processes.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	1	1	1
CO2	2	-	2	1
CO3	2	-	2	3
CO4	1	2	1	1
CO5	3	-	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

E Recommended Study Material:

Text/ Reference Books:

1. S. Chapra and R. Canale, "Numerical Methods for Engineers", McGraw Hill Publication
2. Hoffman J. D. , Taylor and Francis, "Numerical Methods for Engineers and Scientists"
3. Mathews John H. and Fink Kurtis K., "Numerical Methods", Pearson Education
4. Patankar S. V., "Numerical Transfer and Fluid Flow", Hemisphere Publication
5. Date A. W., "Introduction to Computational Fluid Dynamics", Cambridge
6. Donald Hearn and Pauline Baker M., "Computer Graphics", Prentice-Hall of India Pvt. Ltd., 1997.
7. Kunwoo Lee, "Principles of CAD/CAM/CAE Systems", Addison Wesley Pub Co., 1999
8. Chris Mc Mohan, "CAD/CAM: Principles, Practice and Manufacturing", Prentice Hall, 1999
9. Mortenson M. E., "Geometric modeling", Industrial Press.
10. Browne Jim, "Computer Aided Engineering and Design", CIM Research unit, University College, Glway, Ireland.
11. Saxena Anupam and Sahya Birendra, "Computer Aided Engineering Design", Springer
12. Rogers David F. and Adams J. Alan, "Mathematical Elements for Computer Graphics" McGraw Hill, 1990.

Reading Materials, web materials with full citations:

1. http://nptel.iitm.ac.in/
2. Programming Languages: C, C++, MATLAB
3. Software's: AutoCAD, Pro/Engineer, ANSYS, XOR System for Reverse Engineering
4. ASME Journal of Mechanical Design (http://asmedl.aip.org/MechanicalDesign)
5. IEEE Computer Graphics and Applications (http://ieeexplore.ieee.org/servlet/opac?punumber=38)
6. IE Mechanical Engg.
7. www.sciencedirect.com
8. SADHNA (Engineering Science) (http://www.ias.ac.in/sadhana/)

MEUE551: ADVANCED MATERIALS (Programme Elective- I)

M TECH 1st SEM ADVANCED MANUFACTURING TECHNOLOGY

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Composite Materials	18
2.	Thin Film Coatings	13
3.	Testing and Characterization of Advanced Materials	10
4.	Smart Materials	04

Total hours (Theory): 45

Total hours (Practical): 30

Total hours: 75

B Detailed Syllabus:

1	**Composite Materials**	**18 Hours**	**40%**
1.1	Introduction: Definition, Types of matrices and reinforcements, types of composites, properties of composites in comparison with standard materials, applications of metal, polymer, and ceramic matrix composites		
1.2	Manufacturing methods: Polymer Matrix Composites (PMC)- Hand and spray lay-up, vacuum bag method, resin transfer Moulding, compression moulding, filament winding, pultrusion, extrusion, injection moulding.		
1.3	Metal Matrix Composites (MMC): Solid state method, liquid state method, semi-solid state method, vapor deposition, In-situ.		

1.4	Ceramic Matrix Composites (CMC): Chemical vapor or liquid phase infiltration, polymer infiltration and pyrolysis (PIP), hot press sintering		
2	**Thin Film Coatings**	**13 Hours**	**29%**
2.1	Introduction to thin film coatings, fundamentals of thin film deposition		
2.2	Methods for thin film deposition, Advantages and applications of thin films.		
3	**Testing and Characterization of Advanced Materials**	**10 Hours**	**22%**
3.1	Measurement of microstructural, mechanical, thermal, Wettability and optical properties of materials, standards for material testing.		
4	**Smart Materials**	**04 Hours**	**09%**
4.1	Smart Materials: Shape Memory Alloys, Nanomaterials and nanocomposites, Nanorods, Aerogels, Microporous and Mesoporous materials, Inorganic-Organic Composites, Carbon Nanotubes, Modern metallic materials.		

C Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.
- Material testing laboratories visits will be organized for students to explore the facilities. Students are required to prepare a report on visit and submit as a part of the assignment.

D Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Synthesis the composite material for any given engineering application.

CO2: Develop surface coating on various substrate materials to achieve desire properties.

CO3: Characterization and measurement of various properties for given materials.

CO4: Identify suitable materials for given application.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	1	-	1
CO2	2	1	1	2
CO3	2	2	-	2
CO4	1	-	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

E Recommended Study Material:

Text Books:

1. Chawla, Krishan K. Composite materials: science and engineering. Springer Science & Business Media, 2012.
2. Chawla, Krishan K., and Krishan K. Chawla. Metal matrix composites. Springer New York, 1998.
3. Ramdani, Noureddine. Polymer and ceramic composite materials: Emergent properties and applications. CRC Press, 2019.
4. D.M. Mattox, "Handbook of Physical Vapor Deposition (PVD) Processing, Film Formation, Adhesion, Surface Preparation and Contamination Control", Noyes Publications U.S.A., (1998).
5. W.D. Callister, "Fundamentals of Materials Science and Engineering", John Wiley & Sons Inc., New York, (2001).
6. M. Ohring, "The Materials Science of Thin Films", Academic Press Inc, San Diego, (1992).
7. P.M. Martin, "Handbook of Deposition Technologies for Films and Coatings: Science, Applications and Technology", Elsevier USA (2010)

Reference Books:

1. Gay, Daniel, and Suong V. Hoa. Composite materials: design and applications. CRC press, 2007.
2. Thomas, Sabu, et al., eds. Polymer composites, macro-and microcomposites. Vol. 1. John Wiley & Sons, 2012.
3. G. Cao, "Nanostructures & Nanomaterials: Synthesis, Properties & Applications", Imperial College Press London, (2004).

4. A.I. Gusev, A.A. Rempel, “Nanocrystalline Materials”, Cambridge International Science Publishing UK, (2004)
5. B. Bhushan, “Springer Handbook of Nanotechnology”, Springer Berlin Heidelberg New York USA, (2004).
6. H.Y. Erbil, “Surface Chemistry of Solid and Liquid Interfaces”, Blackwell Publishing Ltd, UK, (2006).

Reading Materials, web materials with full citations:

1. S.C. Tjong, H. Chen, “Nanocrystalline materials and coatings”, Materials Science and Engineering R 45 1–88 (2004)
2. B. Duncan, R. Mera, D. Leatherdale, M. Taylor, R. Musgrove, "Techniques for characterising the wetting, coating and spreading of adhesives on surfaces", NPL Report DEPC MPR 020 1-42 (2005).
3. Satyanarayana, V.N.T. Kuchibhatla, A.S.Karakoti, D. Bera, S. Seal, “One dimensional nanostructured materials”, Progress in Materials Science 52 699–913 (2007).

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUE552 : EXPERIMENTAL TECHNIQUES (Programme Elective- I)
M TECH 1st SEM ADVANCED MANUFACTURING TECHNOLOGY

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Basic concepts of Measurement	10
2.	Performance characteristics of Instruments	10
3.	Pressure and Temperature measurement	08
4.	Force, Torque and Strain measurement	08
5.	Vibration & Noise measurement	06
6.	Computer assisted data acquisition, data manipulation, data presentation	03

Total hours (Theory): 45
Total hours (Practical): 30
Total hours: 75

B Detailed Syllabus:

1 Basic concepts of Measurement 10 Hours 30%

1.1 Errors in measurements
1.2 Statistical analysis of data, Regression analysis, correlation, estimation of uncertainty and presentation of data
1.3 Design of experiments

2 Performance characteristics of Instruments 10 Hours 25%

2.1 Introduction
2.2 Static Characteristics and static calibration

2.3	Dynamic Characteristics		
3	**Pressure and temperature measurement**	**08Hours**	**15%**
3.1	Basic method of pressure measurement		
3.2	High pressure measurement		
3.3	Low pressure (Vacuum) measurement		
3.4	Pressure sensors and transducers		
3.5	Calibration of pressure measurements		
3.6	Basic method of temperature measurement and temperature Scales		
3.7	Temperature sensors and transducers		
3.8	Calibration of temperature measurements		
4	**Force, torque and strain measurement**	**08 Hours**	**15%**
4.1	Basic method of force measurement		
4.2	Characteristics of force transducers		
4.3	Force sensors and transducers		
4.4	Torque measurement on rotating shafts		
4.5	Basic method of strain measurement		
4.6	Mechanical and mechanical-optical gauges, Electrical gauges		
4.7	Mounting techniques		
4.8	Temperature compensation		
4.9	Rosettes, Photo elastic method of strain measurement		
5	**Sound and Vibration measurements**	**06 Hours**	**10%**
5.1	Basic method of vibration measurement		
5.2	Vibration sensors and transducers		
5.3	Microphones, Sound level meter, Frequency analyzer		
6	**Computer assisted data acquisition, data manipulation, data presentation**	**03 Hours**	**05%**
6.1	Essential features of data acquisition systems		
6.2	Software compatible with data acquisition systems and simulation		

C Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Analyze the experimental data to do the error analysis and to find the uncertainty in the results.

CO2: Illustrate the various measurement techniques used in the experiments to measure various thermos-physical properties.

CO3: Identify the various intelligent instruments for data acquisition.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	3	-	-	2
CO2	1	2	-	1
CO3	2	1	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

E Recommended Study Material:

Text/ Reference Books:

1. Holman, J.P, "Experimental Methods for Engineers" 5th Ed. McGraw hill International Edition, 1989.
2. Doeblin, E.O.,"Measurement System – Application and Design – McGraw Hill International Ed.,1990
3. Nakra B.C., Chaudhry K. K,"Instrumentation, Measurement and Analysis", 2nd Edition, Tata McGraw-Hill Publishing Company Ltd.,2006.
4. Eckman, D.P. "Industrial Instrumentation", Wiley Eastern Ltd., New Delhi, 1990
5. Hale, J. and Kocak, H., "Dynamics and Bifurcations", Springer-Verlag, N.Y. 1991

6. Strogatz, S.H., “Nonlinear Dynamics and Chaos”, Addison Wesley, Massachusetts, 1995.
7. Helfrack, A.D. and Cooper, W.D., “Modern Electronic Instrumentation & Measurement Techniques”, Prentice Hall of India Pvt. Ltd., New Delhi -2001.

Reading Materials, web materials with full citations:

1. http://www.sciencedirect.com/science/journal
2. http://www.ieindia.org/publish/pr/pr.htm
3. http://www.ieindia.info/public.asp/me/me.htm
4. http://www.ias.ac.in/sadhana

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to Industry 4.0	05
2.	Main components of the industry 4.0	10
3.	Industry 4.0 design principles	10
4.	Smart factory	10
5.	Standards of industry 4.0 and Safety and Security in networked Production Environments	10

Total hours (Theory): 45
Total hours (Practical): 30
Total hours: 75

B Detailed Syllabus:

1 Introduction to Industry 4.0 **05 Hours 12%**

1.1 Industry 1.0 TO 3.0

1.2 Meaning of industry 4.0

1.3 Comparison of Industry 4.0 Factory and today's Factory

1.4 Difference between conventional automation and Industry 4.0

2 Main components of the industry 4.0 **10 Hours 22%**

2.1 Cyber-Physical Systems

2.2 Internet of Things

2.3 Smart Factory

2.4 Internet of services

3	**Industry 4.0 design principles**	**10 Hours**	**22%**
3.1	Horizontal integration		
3.2	Vertical integration		
4	**Smart factory**	**10 Hours**	**22%**
4.1	Steps to a smart factory		
4.2	Basic principles and technologies of a Smart Factory		
5	**Standards of industry 4.0 and Safety and Security in networked Production Environments**	**10 Hours**	**22%**
5.1	Success with standards		
5.2	Road map of standards		
5.3	Case studies approaching industry 4.0		
5.4	Safety for connected Machines and Systems		
5.5	Safety in Human Robot cooperation		

C Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Explain the concept of internet of Things with respect to earlier revolutions.

CO2: Identify the main components of industry 4.0.

CO3: Explain design principles of industry 4.0 and internet of things.

CO4: Explain the basic principles for industry 4.0.

CO5: Explain various standards and safety issues of Industry 4.0

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	3	-	-	2
CO2	1	2	-	1
CO3	2	1	-	1
CO4	2	1	-	1
CO5	2	1	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

E Recommended Study Material:

Text/ Reference Books:

1. Gilchrist, Alasdair "Industry 4.0,", 1st Edition.2016
2. Madisetti, Arshdeep Bahgavijay, "Ïnternet of Things,", 1st Edition.2014
3. Dr. SRN Reddy, Rachit Thukral and Manasi Mishra, "Introduction to Internet of Things: A practical Approach", ETI Labs
4. Cuno Pfister, "Getting Started with the Internet of Things", O Reilly Media Maker Media, Inc; 1st edition 2011.
5. Alp Ustundag and Emre Cevikcan,"Industry 4.0: Managing the Digital Transformation".
6. Bartodziej, Christoph Jan,"The Concept Industry 4.0".

Reading Materials, web materials with full citations:

1. https://www.nist.gov/topics/manufacturing
2. https://www.i-scoop.eu/industry-4-0/

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING

MEUE554 : MANUFACTURING PLANNING AND CONTROL
(Programme Elective-II)

M TECH 1st SEM ADVANCED MANUFACTURING TECHNOLOGY

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to Manufacturing Systems Engineering	06
2.	Resource Planning & Production Control	10
3.	Just in Time (JST) Production	06
4.	Toyota Production System (TPS)	06
5.	Ergonomics, DFM, QFD	06
6.	Experimental Design: Statistical Analysis of Data	06
7.	Methodologies for Improving OEE in Manufacturing	05

Total hours (Theory): 45
Total hours (Practical): 30
Total hours: 75

B Detailed Syllabus:

1 Introduction to Manufacturing Systems Engineering 06 Hours 13%

1.1 Role and scope for manufacturing planning and control.
1.2 Essentials steps in control activity.
1.3 Elements of production control.
1.4 Checklist of information required for production planning and control.

2	**Resource Planning & Production Control**	**10 Hours**	**23%**
2.1	Six stages of factory planning.		
2.2	System engineering and factory planning.		
2.3	Materials requirements planning, evolution from MRP to MRP II.		
2.4	Computer aided planning and control.		
3	**Just in Time (JIT) Production**	**06 Hours**	**13%**
3.1	Introduction- the spread of JIT movement, some definitions of JIT.		
3.2	Core Japanese practices of JIT, profit through cost reduction.		
3.3	Elimination of over production.		
3.4	Quality control, quality assurance.		
3.5	Respect for humanity, Flexible work force.		
3.6	JIT production adapting to changing production quantities.		
3.7	Process layout for shortened lead times.		
3.8	Standardization of operation, automation.		
4	**Toyota Production System (TPS)**	**06 Hours**	**13%**
4.1	Philosophy of TPS, basic frame work		
4.2	Kanbans, determining number of Kanbans inTPS (a) Kanban number under constant quantity withdrawal system. (b) Constant cycle, non-constant quantity withdrawal system. (c) Constant withdrawal cycle system for the supplier Kanban.		
4.3	Supplier Kanban and the sequence schedule for use by suppliers (a) Later replenishment system by Kanban. (b) Sequence withdrawal system.		
4.4	Production smoothing in TPS, production planning, production smoothing, adaptability to demand fluctuations.		
4.5	Sequencing method for the mixed model assembly line to realize smoothed production of goal.		
5	**Ergonomics, DFM, QFD**	**06 Hours**	**13%**
5.1	Ergonomics: Human factors considerations in design and applications.		

5.2 DFM: Design for manufacturing – including assembly aspects.

5.3 DFE: Design for Environment- environmental objectives & global issues, drilling and other

5.4 QFD-CE: Quality function deployment – concurrent engineering.

5.5 PDD-PC: Product design and development-product cycle.

6 Experimental Design: Statistical Analysis of Data **06 Hours 13%**

6.1 Purpose of Statistical Analysis

6.2 Descriptive Statistics

6.3 Inferential Statistics

6.4 Use of Statistical Software

7 Methodologies for Improving OEE in Manufacturing **05 Hours 12%**

7.1 Lean Manufacturing

7.2 Six Sigma

7.3 Theory of Constraints

C Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behaviour will be observed strictly.

D Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Modify the components and characteristics of manufacturing systems.

CO2: Test appropriate performance of Resource planning pertaining to material and manufacturing resource planning.

CO3: Criticize the performance of manufacturing system as compared with lean principles along with use of basic framework of Toyota production system

CO4: Relate current product and process in context with environmental objectives & global issues by identifying various global standards which industry update time to time.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	1	1	1	2
CO2	1	2	2	1
CO3	3	2	3	3
CO4	1	2	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

E Recommended Study Material:

Text Books:

1 Roberta S. Russell, Bertrand Russell & Bernard W. Taylor, Production and Operations Management, Pearson Higher Education & Professional Group, 1995

2 P. B. Mahapatra, "Operations management: a quantitative approach", PHI, 2010

3 R.L.Francis andJ.A. White – Facility Layout and Location, Prentice Hall Inc.1992

4 Principles of Process Planning G. Halevi and R.D. Weill Chapman and Hall,1995

5 Production and Operations Management S.N. Chary Tata McGraw Hill

6 Operations Management J.G. Monks McGraw Hill

7 Facility Layout and Location Francis and White Prentice Hall

09 Jhamb L C, A text book of production (operations) management, Everest Publishing House, 7TH ED, 2002

10 Mahajan M S, Industrial Engineering and Production Management, Dhanpat Rai and co. Limited, New Delhi, 3rd Edition 2010

11 Monks Joseph G, Schaum's outline of theory and problems of operations management, McGraw-Hill Professional 5th Edition (2004)

12 Panneerselvam R, Production and Operations Management, PHI Learning Pvt. Ltd., 2006

Reference Books:

1 James P.Womack" Lean Thinking"Freepres 2003.

2 Alavudeen & N. Venkateshwaran, "Computer integrated manufacturing", PHI, 2005

3 Chris McMahon & Jimmie Browne, "CAD CAM principles, practice and manufacturing management", Pearson Education, 2000

4 Joseph S & Martinich P, Production And Operations Management: An Applied Modern Approach, Wiley-India, 2008
5 Karl. T. Ulrich, Steven D. Eppinger, “Product design and development”, McGraw Hill, 2000
6 A. C. Chitale and R. C. Gupta, “Product design and manufacturing”, PHI
7 Timjones Butterworth Heinmann, “New product development”, Oxford, UCI, 1997
8 Chary S N, Production and Operations Management, Tata McGraw Hill, 3rd Ed. 2004
9 Krishnaswamy & Mathirajan, Cases In Operations Management, PHI Learning Pvt. Ltd. 2001
10 Geoffery Boothroyd, Peter Dewhurst and Winston Knight, “Product design for manufacture and assembly”
11 Dieter, George E., “Engineering design - a materials and processing approach”, McGraw Hill, 2000
12 Askin R G and Goldberg J B (2002), Design and Analysis of Lean Production Systems, John Wiley and Sons.

Reading Materials, web materials with full citations:

1. Journal of Indian Institution of Industrial Engineering (http://www.iiie-india.com)
2. Mechanical Engg. (Inst. of Engineers) http://www.ieindia.org/publish/mc/mc.htm
   Production Engg. (Inst. of Engineers) http://www.ieindia.org/publish/pr/pr.htm
3. Manufacturing Technology & Management : Quarterly Journal of the Indian Institution of Production Engineers
5. UDYOG PRAGATI: The Journal for Practicing Managers National Institute of Industrial Engineering (NITIE)
6. Mini Tab 16.0.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUE555: ADVANCED METROLOGY (Programme Elective- II)
M TECH 1st SEM ADVANCED MANUFACTURING TECHNOLOGY

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Metrology Principles and Structures	08
2.	Metrology Strategy and Quality	12
3.	Surface Topography Analysis	09
4.	Strength and Wear measurement	08
5.	Elasticity and hardness measurements	08

Total hours (Theory): 45

Total hours (Practical): 30

Total hours: 75

B Detailed Syllabus:

1 Metrology principles and structures 08 Hours 18%

1.1 Metrology: introduction, international bureau of weights and measures, quantum standards: a metrological revolution, regional metrology organizations.

1.2 Mutual Recognition of National Metrology Institute (NMI) standards: The International Committee of Weights and Measures (CIPM) and Mutual Recognition Arrangement (MRA), metrology in the 21st century, the SI system and new science.

2 Metrology strategy and quality 12 Hours 26%

2.1 Traceability of measurements: introduction, importance, terminology, traceability of measurement results to SI units, calibration of measuring and testing devices.

2.2 Inter-laboratory Comparisons (ILC) and Proficiency Testing (PT): the benefit of participation in PT, selection of providers and sources of information, causes of unsatisfactory performance.

2.3 Certified reference materials: introduction and definitions, classifications, activities of international organizations such as standardization bodies, accreditation bodies, metrology organizations and user organizations, reference procedures.

2.4 Accreditation and peer assessment: accreditation of conformity, assessment bodies, measurement competence: assessment and confirmation, peer assessment schemes.

2.5 Computer aided inspection

3 Surface topography analysis 09 Hours 20%

3.1 Introduction, measurement and filtering, visualization, quantification of surface texture, conventional 2-D parameters, other parameters defined by ISO, 3-D parameters.

3.2 Optical techniques: optical stylus profilometry, confocal microscopy, interference microscopy.

3.3 Scanning probe microscopy: introduction, atomic force microscopy operation modes, applications and limitations of surface measurement.

4 Strength and wear measurement 08 Hours 18%

4.1 Wear: introduction, types of wear, units for wear, approach to tribological testing, test parameters, interaction with other degradation mechanisms, quantitative assessment of wear such as dimensional change, mass and volume loss, etc.

4.2 Friction force measurements, strength analysis: introduction, interface strength: adhesion measurement methods, scratch test for thin-film coatings.

5 Elasticity and hardness measurements 08 Hours 18%

5.1 Elasticity: introduction, development of elasticity theory, stress, strain and elastic constants, instrumented indentation as a method of determining elastic constants, determining the elastic modulus by indentation.

5.2 Hardness: introduction, Instrumented Indentation Test (IIT), Martens Hardness (HM) determination of martens hardness, nanoindentation method, hardness test and strength/hardness relations.

C Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.

- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.
- Material testing laboratories visits will be organized for students to explore the facilities. Students are required to prepare a report on visit and submit as a part of the assignment.

D Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Analysis Surface Topography of materials.

CO2: Measure the Strength and wear of various materials.

CO3: Identify the Metrology Strategy and Quality.

CO4: Measure the hardness of materials.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	-	-	1
CO2	2	1	-	2
CO3	1	-	-	1
CO4	2	1	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E Recommended Study Material:

Text Books/ Reference Book:

1. Horst Czichos, Tetsuya Saito, Leslie Smith, “Springer Handbook of Materials Measurement Methods”, Springer Science Business Media Inc., (2006), Germany.
2. John G. Webster, “Measurement, Instrumentation, and Sensors Handbook”, Taylor & Francis CRC Press LLC, (1999), USA.
3. Bharat Bhushan, “Springer Handbook of Nanotechnology”, Springer Science Business Media Inc., (2006), New York

4. Horst Czichos, Tetsuya Saito, Leslie Smith, "Springer Handbook of Metrology and Testing", Springer Science Business Media Inc., 2nd edition, Germany.
5. International Organization for Standardization (ISO): International vocabulary of basic and general terms in metrology (BIPM, IEC, IFCC, ISO, IUPAC, IUPAP, OIML, Genf (1993).
6. A. J. Wallard, T. J. Quinn, "Intrinsic standards-are they really what they claim?", Cal Lab Magazine, Vol. 6, No. 6, Nov./Dec. (1999).
7. W. Martienssen, H. Warlimont (Eds.): "Springer Handbook of Condensed Matter and Materials Data", Springer, (2005), Berlin, Heidelberg.
8. U. Kurfurst, A. Desaules, A. Rehnert, H. Muntau, "Estimation of measurement uncertainty by the budget approach for heavy metal content in soil under different land use", Accred. Qual. Assur. 9, 64-75 (2004).
9. Analytical Methods Committee (AMC), "Uncertainty of Measurement: Implications of its Use in Analytical Science", Analyst, 120, 2303-2308 (1995)
10. M. H. Ramsey, S. Squire, M. J. Gardner, "Synthetic reference sampling target for the estimation of measurement uncertainty", Analyst 124(11), 1701-1706 (1999).
11. S. Squire, M. H. Ramsey, M. J. Gardner, D. Lister: "Sampling proficiency test for the estimation of uncertainty in the spatial delineation of contamination", Analyst 125(11), 2026-2031 (2000)

Reading Materials, web materials with full citations:

1. http://nptel.iitm.ac.in/
2. http://www.sciencedirect.com/science/journal
3. http://www.ieindia.info/public.asp/me/me.htm
4. http://www.ias.ac.in/sadhana

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
HSUS502 : FOREIGN LANGUAGES (French)
M TECH 2nd SEM ADVANCED MANUFACTURING TECHNOLOGY

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	--	02	30	02
Marks	--	100	100	

A Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to French Language	08
2.	Grammar: Articles, Tense, Forms, Numbers, Verbs, Days, Months, Family	08
3.	Grammar : Adjectives, Adverbs, Interrogative Forms, Directions, Countries, Nationalities, Seasons, Weather, Professions, Verbs	08
4.	Grammar: Prepositions, Conjunctions, Tenses, Colours, Vegetables, Fruits, Shapes, Verbs	06

Total hours (Theory): 00

Total hours (Practical): 30

Total hours: 30

B Detailed Syllabus:

1.	Introduction to French Language	08 Hours	28%
	Facts and figures about French Language; Basic French Linguistics-* Alphabets * Accents * Liaison * Nasalization French Culture, Differ between French and English; **Grammar-** Subject Pronoun, Verbs: (être, avoir, habiter, regarder, manger … "er" verb), Form of address, Numbers (1 to 20), Nouns and plurals of nouns, The expression: C'est, Il y a; **Presentation:** -1) Self-Introduction-2) Question and answering; **Dialogue**		
2.	Grammar: Articles, Tense, Forms, Numbers, Verbs, Days, Months, Family	08 Hours	28%

	Grammar-Definite articles**,** Indefinite articles**,** Present tense (Positive Forms**,** Negative Forms)**,** Numbers (21 to 100, 100-1000)**,** Days, Months, Family, Verbs: (aller, venir, finir, pouvoir, vouloir … "ir" verb)**; Social Links-**1), My family & relations 2) Appointments 3) Gathering information from someone**; Dialogue**		
3.	Grammar : Adjectives, Adverbs, Interrogative Forms, Directions, Countries, Nationalities, Seasons, Weather, Professions, Verbs	08 Hours	28%
	Grammar- Common Adjectives**,** Comparative Adjectives**,** Common Adverbs**,** Interrogative Forms**,** The expression: "On"**,** Directions, Countries, Nationalities, Seasons, Weather, Professions**,** Verbs: (Prendre, Apprendre, Comprendre, faire ... "re " verb)**; Work , Study and Travel-**1) Job/ Profession 2) Ticket Reservation (At Bus/At Railway/At Airport)**; Dialogue**		
4.	Grammar: Prepositions, Conjunctions, Tenses, Colours, Vegetables, Fruits, Shapes, Verbs	06 Hours	26%
	Grammar-1) Common Prepositions 2) Common Conjunctions 3)Past Tense 4) Future Tense 5) Colors ,Shapes, Animals ,Vegetables, Fruits 6) Verbs: ("er", "ir","re" etc...)**; Food & Shopping-**1) Buy a vegetables and fruits 2) Any Conversation between Customer and Vendor (At Mall/At Restaurant / At Market)**; Dialogue**		

C Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.
- Material testing laboratories visits will be organized for students to explore the facilities. Students are required to prepare a report on visit and submit as a part of the assignment.

D Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Gain basic communication skills in French language with preliminary understanding of grammar.

CO2: Develop vocabulary required to speak about him/herself and his/her immediate environment.

CO3: Become capable of interacting in simple ways, to ask simple questions to get necessary information, to reply simple questions.

CO4: Become capable of understanding and using simple instructions in their personal, academic and professional environments.

CO5: Develop skills and intelligences to function in multi-disciplinary and cross-cultural work environment.

CO6: Practice new global trends in communication in multiple perspectives at personal, professional, and social level.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	-	-	2	-
CO2	-	-	2	-
CO3	-	-	2	-
CO4	-	-	2	-
CO5	-	-	2	-
CO6	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

E Recommended Study Material:

Text Books/ Reference Book:

1. Complete French: All-In-One, McGraw-Hill, Amazon
2. Best for Grammar: Easy French Step-by-Step, McGraw-Hill, Amazon
3. Basic French: McGraw-Hill, Amazon
4. French Grammar for Beginners, Amazon

Reading Materials, web materials with full citations:

1. https://alison.com/course/french-language-studies-introduction
2. https://alison.com/course/basic-french-language-skills-for-everyday-life-revised-2017
3. http://www.bbc.co.uk/languages/french/
4. https://www.loecsen.com/en/learn-french

5. https://www.youtube.com/watch?v=ujDtm0hZyII
6. https://alison.com/course/french-language-studies-introduction

HSUS501 : ACADEMIC SPEAKING AND PRESENTATION SKILLS
M TECH 2nd SEM ADVANCED MANUFACTURING TECHNOLOGY

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	--	02	30	02
Marks	--	100	100	

A Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Foundations of Advance Communication	04
2.	Art of Conversation	06
3.	Science of Power Speaking	06
4.	Academic Speaking Application – Part I	08
5.	Academic Speaking Application – Part II	06

Total hours (Theory): 00
Total hours (Practical): 30
Total hours: 30

B Detailed Syllabus:

1.	**Foundations of Advance Communication**	04 Hours	14%
	Meaning and Definition of Advance Communication; Advance Communication in Digital, Social, Mobile World; Strategies for Advance Communication; Meaning and Concept of Academic Language; High Frequency Academic Vocabulary		
2.	**Art of Conversation**	06 Hours	20%
	Describing people, places and things; Expressing opinions; Making suggestions; Persuading someone; Interpreting and Summarizing		
3.	**Science of Power Speaking**	06 Hours	20%

	Phonemes, Word Stress, Pronunciation, Intonation, Pause, Register, Fluency, Prosody, Lexical Range		
4.	**Academic Speaking Application – Part I**	08 Hours	26%
	Art of Oratory, Formal Presentation, Speech Analysis – Decoding Best Speeches		
5.	**Academic Speaking Application – Part II**	06 Hours	20%
	Job Interview, Group Discussion, Meeting		

C Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.
- Material testing laboratories visits will be organized for students to explore the facilities. Students are required to prepare a report on visit and submit as a part of the assignment.

D Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Understand and demonstrate advance communication and academic speaking skills

CO2: Demonstrate ability to communicate in diverse situations

CO3: Activate and extend their linguistic and communicative competence

CO4: Demonstrate the formal presentation skills

CO5: Demonstrate performing ability at group discussion and personal interview

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	-	-	2	-
CO2	-	-	2	-
CO3	-	-	2	-
CO4	-	-	1	-
CO5	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

E Recommended Study Material:

Text Books/ Reference Book:

1. Business Communication Today (Thirteenth Edition) by Courtland L. Bovee, John V. Thill and Roshan Lal Raina
2. Effective Speaking Skills by Terry O' Brien
3. Speak Better Write Better by Norman Lewis
4. Well Spoken: Teaching Speaking to All Students by Erik Palmer
5. Let Us Hear Them Speak : Developing Speaking – Listening Skills in English by Jayshree Mohanraj (Publisher – Sage Publication)
6. The craft of scientific presentations: Critical steps to succeed and critical errors to avoid. New York: Springer by Michael Alley
7. Presentation Skills in English by Bob Dignen (Publisher: Orient Black Swan)

Reading Materials, web materials with full citations:

1. TED Talk : How to speak so that people want to listen
https://www.ted.com/talks/julian_treasure_how_to_speak_so_that_people_want_to_listen?language=en
2. TED Talk: The 110 techniques of communication and public speaking
https://www.ted.com/talks/david_jp_phillips_the_110_techniques_of_communication_and_public_speaking

M. Tech. (AMT) Programme

SYLLABI
(Semester – 2)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUC554: MACHINING AND FORMING PROCESSES
M TECH 2nd SEM ADVANCED MANUFACTURING TECHNOLOGY

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

A Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to machining	04
2.	Analysis of metal cutting process	10
3.	Machinability and tool life	10
4.	Machining dynamics	06
5.	Metal forming	15
6.	Analysis of metal forming process	10
7.	Computer analysis of machining and metal forming processes	05

Total hours (Theory): 60

Total hours (Practical): 30

Total hours: 90

B Detailed Syllabus:

1 Introduction to machining **04 Hours** **08%**

1.1 Single and multi-point cutting tools, Tool geometry, Conventional machining processes.

1.2 Elastic and plastic behavior during machining of metals.

1.3 Orthogonal and oblique cutting.

2	**Analysis of metal cutting process**	**10 Hours**	**16%**
2.1	Determination of cutting and thrust forces, Dynamometer, Measurement of shear angle shear strain, Velocity relationship.		
2.2	Merchant circle, Chip formation analysis.		
2.3	Friction during metal cutting, Heat generation and dissipation during machining, Tool-work temperature measurement techniques.		
3	**Machinability and tool life**	**10 Hours**	**16%**
3.1	Machinability measurement criteria.		
3.2	Tool wear mechanisms - progressive wear, flank wear and crater wear.		
3.3	Prediction and monitoring of tool wear, Tool fracture, Tool failure - thermal cracking & softening, mechanical chipping.		
3.4	Mechanism of wear – abrasion, adhesion, diffusion, chemical wear, Measurement and optimization of tool life, Factors affecting tool life.		
4	**Machining dynamics**	**06 Hours**	**10%**
4.1	Types of machine tool vibrations.		
4.2	Sources of vibrations, Effect of vibrations, Vibration control.		
5	**Metal forming**	**15 Hours**	**24%**
5.1	Fundamentals of metal forming processes, Conventional forming processes, relation to material science, Theories of plasticity and mechanics of plastic deformation.		
5.2	Stress - strain relationships, deformation equations, work hardening and yield criteria for ductile metals.		
5.3	Cold, hot and warm forming.		
5.4	Stretch forming, Spinning, Unconventional or High energy rate forming (HERF) process - Explosive forming, Rubber forming or hydroforming, Magnetic pulse or Electro-magnetic forming, Electro-hydraulic forming, Blow or vacuum forming, Thermo forming.		
5.5	L-SAW & spiral-SAW pipe forming processes.		
6	**Analysis of metal forming process**	**10 Hours**	**16%**
6.1	Anisotropy in sheet metal, Effect of temperature and strain rate during forming.		
6.2	Formability and forming limit diagram.		

6.3 Formability tests – Olsen & Erichsen test, swift cup test, Fukui test.

6.4 Slab method and Slip line theory.

7 Computer analysis of machining and metal forming processes 05 Hours 10%

7.1 Analysis of machining & forming processes, Simulation.

7.2 Various software, Advantages of computer analysis.

C Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.
- Industrial visits will be organized for students to explore industrial facilities. Students are required to prepare a report on industrial visit and submit as a part of the assignment.

D Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Understand the theory behind various machining and forming processes and can implement the knowledge to analyze various metal operations.

CO2: Aware about machinability and dynamics of machining.

CO3: Familiar with the simulation of machining and metal forming process.

CO4: Get practical with the current trends in industries and update themselves with the latest technologies.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	1	1	2	1
CO2	2	3	1	1
CO3	3	1	2	2
CO4	2	2	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

E Recommended Study Material:

Text Books/ Reference Book:

1. G. Boothroyd and W.A. Knight, "Fundamentals of Machining and Machine Tools", CRC press.
2. A. Ghosh and A.K. Mallik, "Manufacturing Science", Affiliated East-West Press (Pvt.) Ltd.
3. A Bhattacharya, "Metal Cutting: Theory and Practice", Central Book Publishers.
4. J.A. Schey, "Introduction to Manufacturing Processes", McGraw Hill.
5. G. W. Rowe, "Principles of Industrial Metalworking Processes", CBS Publishers.
6. Surender Kumar, "Principles of Metal Working", Oxford & IBH Publishing Company
7. Milton C. Shaw, "Metal Cutting Principles", CBS Publishers.
8. Edward M. Trent, "Metal Cutting", Elsevier.
9. B. Avitzer, "Metal Forming: Processes and Analysis", Tata McGraw Hill.
10. W.F. Hosford and R.M. Caddell, "Metal Forming: Mechanics and Metallurgy" Cambridge University Press.
11. Warren R. DeVries, "Analysis of Material Removal Processes", Springer.

Reading Materials, web materials with full citations:

1. http://nptel.iitm.ac.in/courses/Webcoursecontents/IIT%20Kharagpur
2. Manuf%20Proc%20II/New_index1.html
3. http://www.sciencedirect.com/science/journal
4. http://www.ieindia.org/publish/pr/pr.htm
5. http://www.ieindia.info/public.asp/me/me.htm
6. http://www.ias.ac.in/sadhana

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUC555 : CNC TECHNOLOGY & PROGRAMMING
M TECH 2nd SEM ADVANCED MANUFACTURING TECHNOLOGY

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	00	100	

A Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Fundamentals of NC/CNC machine tools	04
2.	Basics of CNC machine tools	08
3.	Basics of CNC programming	05
4.	Basics of turning center programming	08
5.	Basics of machining center programming	08
6.	Sub programming & macro programming	04
7.	Computer aided CNC part programming	08

Total hours (Theory): 45

Total hours (Practical): 00

Total hours: 45

B Detailed Syllabus:

1 Fundamentals of NC/CNC machine tools 04 Hours 10%

1.1 Introduction to NC, CNC & DNC systems.

1.2 Comparison of NC, CNC & conventional machine tools.

1.3 Advantages & limitations of NC/CNC machine tools.

1.4 Role of NC/CNC technology in modern manufacturing.

2	**Basics of CNC machine tools**	**08 Hours**	**17%**
2.1	Axes designation, reference points & CNC control system		
2.2	Automatic tool changer & automatic pallet changer		
2.3	Machine tool structure, guideways & transmission system		
2.4	Drives and feedback devices, NC/CNC tooling		
3	**Basics of CNC programming**	**05 Hours**	**12%**
3.1	Coding systems & types of codes		
3.2	Types of programming & programming functions		
3.3	Structure of CNC part program		
4	**Basics of turning center programming**	**08 Hours**	**17%**
4.1	Preparatory functions & tool compensations		
4.2	Automatic reference point return & dwell		
4.3	Single pass & multipass canned cycles		
5	**Basics of machining center programming**	**08 Hours**	**17%**
5.1	Preparatory functions, profile milling & cutter compensations		
5.2	Drilling, tapping & boring canned cycles		
6	**Subprogramming &Macro programming**	**04 Hours**	**10%**
6.1	Pattern repeating, mirroring, rotation & scaling		
6.2	Applications of macros, variables & looping		
7	**Computer aided CNC part programming**	**08 Hours**	**17%**
7.1	Need for computer aided part programming.		
7.2	Tools for computer aided part programming.		
7.3	APT programming, language features & APT statements.		
7.4	CAD/CAM based part programming & evaluation of CAM software.		

C Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.

- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behaviour will be observed strictly.
- Industrial visits will be organized for students to explore industrial facilities. Students are required to prepare a report on industrial visit and submit as a part of the assignment.

D Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Explain advanced features of NC/CNC technology and programming.

CO2: Develop NC/CNC part program for turning and machining center.

CO3: Apply sub-programming and canned cycles for quicker & easier development of part program.

CO4: Use computer aided manufacturing software to create part program.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	1	-	1	-
CO2	3	-	2	3
CO3	2	1	1	2
CO4	1	-	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E Recommended Study Material:

Text Books/ Reference Book:

1. T. K. Kundra, P. N. Rao and N. K.Tewari, Numerical control and computer aided manufacturing., Tata McGraw Hill Publishing company Ltd.
2. Yoram Koren and Joseph Ben-Uri, Numerical Control of Machine Tools, Khanna Publishers, Delhi.
3. P. Radhakrishnan, Computer numerical control machines, New Central Book Agency.
4. P. Radhakrishan, S. Subramaniyam, CAD, CAM and CIM, New Age International.
5. Mattson Mike & Mazidi J G , CNC Programming Principles and Applications, Thomson Brooks.
6. Sinha S K, CNC Programming, Galgotia Publications.
7. Steave Krar & Arthur Gill, CNC Technology and Programming, McGraw–Hill Publishing.

8. P M Agrawal and V J Patel, CNC Fundamentals and programming, Charotar Pub.

Reading Materials, web materials with full citations:

1. http://nptel.iitm.ac.in/
2. Software’s: Master CAM, Pro/Engineer
3. www.sciencedirect.com
4. Mechanical Engg. (Inst. of Engineers) http://www.ieindia.org/publish/mc/mc.htm
   Production Engg. (Inst. of Engineers) http://www.ieindia.org/publish/pr/pr.htm
5. IEEE transactions on Manufacturing Technology
   http://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=8218
6. IET Manufacturing Engineer http://ieeexplore.ieee.org/servlet/opac?punumber=2189
7. http://www.ias.ac.in/sadhana

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUC556 : INDUSTRIAL AUTOMATION & ROBOTICS
M TECH 2nd SEM ADVANCED MANUFACTURING TECHNOLOGY

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	--	3	3
Marks	100	--	100	

A Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to Automation	02
2.	Automation Tools	11
3.	Machine Vision	04
4.	Hydraulic Control in Automation	05
5.	Fundamentals of Industrial Robots	02
6.	Robotic Control System, End Effectors and Sensors	07
7.	Robot Programming	03
8.	Robot Arm Kinematics and Dynamics	11

Total hours (Theory): 45

Total hours: 45

B Detailed Syllabus:

1 Introduction to automation 02 Hours 04%

1.1 Introduction, applications, goals, issues of automation

1.2 Advantages, problems of automation, Low cost automation

2 Automation tools 11 Hours 25%

2.1 Hardware components for automation like sensors, actuators and data acquisition systems

2.2 PID controllers, programmable logic controller, programmable automation controllers

2.3 Distributed control system, human machine interface, supervisory control.

3	**Machine vision**	**04 Hours**	**08%**
3.1	Introduction and applications		
3.2	Components of a machine vision system		
3.3	Image acquisition and representation		
4	**Hydraulic control in automation**	**05 Hours**	**12%**
4.1	Introduction and principles of fluid power transmission		
4.2	Applications of fluid power in industrial automation		
4.3	Study of elements of hydraulic system		
4.4	Design, analysis and study of typical hydraulic systems		
5	**Fundamental of Industrial Robotics**	**02 Hours**	**04%**
5.1	Specifications, Characteristics, Components, Configurations, Selection criteria and Applications		
6	**Robotic control system, end effectors and sensors**	**07 Hours**	**15%**
6.1	Drives, robot motions, actuators, power transmission systems		
6.2	Robot controllers and dynamic properties of robots, transducers and sensors		
6.3	End effectors, active and passive elements		
7	**Robot programming**	**03 Hours**	**07%**
7.1	Methods of programming		
7.2	Study of various programming languages		
8	**Robot arm kinematics and dynamics**	**11 Hours**	**25%**
8.1	Direct kinematics and solutions		
8.2	Inverse kinematics and solutions		
8.3	Robot arm dynamics, lagrange and newton euler formulations		
8.4	Generalised D’Alembert’s equation of motion		

C Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.

- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behavior will be observed strictly.
- Students will visit various industries/research centers/university departments in the vicinity to observe the manufacturing setups/softwares.

D Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Use the different control system for industrial automation.

CO2: Interpret the different components of machine vision.

CO3: Solve the problem of automation using pneumatic or hydraulic control.

CO4: Analyse the effector or gripper used in robotics.

CO5: Evaluate robotic motion by solving kinematics and dynamics of robot arm manipulator with theory and some software.

CO6: Prepare the program for motion of robot arm.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	1	-	1
CO2	2	-	-	1
CO3	3	-	1	3
CO4	1	-	1	1
CO5	3	2	2	3
CO6	1	1	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

E Recommended Study Material:

Text Books/ Reference Book:

1. Groover M P, "Automation, Production Systems and Computer Integrated Manufacturing", Third Edition Pearson Educational Publication
2. Boothroyd Geoffry, "Assembly Automation and product design", Second Edition, CRC Press
3. Khaimoswitch E M, "Hydraulic control of machine tools", Pergamon press

4. Gupta A K and Arora S K, “Industrial Automation and Robotics”, Third Edition, Laxmi Publication Ltd.
5. Klafter R D, Chmielewski T A and Negin M, “Robot Engineering:An Integrated approach”,PHI
6. Shahinpoor Moshen, “A Robot Engg Text Book”, Harper and Row Publishers, NY.
7. Bolton W, “Programmable Logic Controllers”, Fourth edition, Elsevier Publication, UK.
8. Pippenger John and Tyler Gregory Hicks, “Industrial hydraulics”, Third edition, McGraw-Hill, 1979.
9. Groover and Zimmer, “CAD/CAM: computer-aided design and manufacturing”, Fifth edition Prentice-Hall.
10. Thomas H A, “Handbook of low cost automation techniques”, Gower, 1969
11. CMTI, “Machine tool design”, Tata McGraw hill publication
12 Groover M P, “Industrial Robotics Technology, programming and applications”, McGraw-Hill
13. Schilling Robert J, “Fundamentals of Robotics, Analysis and Control”, Prentice Hall of India
14. Craig John J, “Introduction to Robotics, Mechanics and control”, 3rd Ed, Addison – Wesley
15. David Vernon, “Machine Vision”, Prentice Hall, New York.
16. Mittal R K and Nagrath I J, “Robotics and Control”, McGraw-Hill, 2013.

Reading Materials, web materials with full citations:

1. http://nptel.iitm.ac.in
2. Simbad 3D, Robo Works 3.0, Lab View 2010
3. Production Engg. (Inst. of Engineers)- http://www.ieindia.org/publish/pr/pr.htm
4. Journal of Mechanisms and Robotics (ASME)- http://asmedl.aip.org/JMR
5. Journal of Robotics and Automation (IEEE) http://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=56
6. Sadhna (http://www.ias.ac.in/sadhana/)
7. Mechanical Engg. (Inst. of Engineers) http://www.ieindia.org/publish/mc/mc.htm
8. IEEE Transactions on Automation Science and Engineering (http://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=8856)
9. IET Control and Automation (http://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=4469873)

MEUS551 : AUTOMATION LABORATORY
M TECH 2nd SEM ADVANCED MANUFACTURING TECHNOLOGY

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	02	02	01
Marks	-	50	50	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	CNC Programming for Milling Center	06
2.	CNC Programming for Turning Center	06
3.	Computer Aided CNC Programming	04
4.	Computer Aided Inspection	02
5.	Reverse Engineering	02
6.	Rapid Prototyping	02
7.	Hydraulic & Pneumatic Control in Automation	04
8.	Robot Programming	04

Total Hours (Lab): 30

Total Hours: 30

B. Instructional Method and Pedagogy:

- Laboratory study involves demonstration/hands-on practice in relevance of course content.
- Attendance is compulsory in laboratory.
- Assignments based on course content will be given to the students.

- Experiments/Demonstration/Tutorials related to course content will be carried out in the laboratory.
- In the laboratory discipline and behavior will be observed strictly.

C. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Prepare a program for CNC machine by using software and manually.

CO2: Solve the problem of automation using pneumatic or hydraulic control.

CO3: Create the program for motion of robot arm using software.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	1	2	2
CO2	2	1	2	2
CO3	3	1	1	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

D. Recommended Study Material:

Reference Books:

1. Boothroyd Geoffry, "Assembly Automation and product design", Second Edition, CRC Press
2. Khaimoswitch E M, "Hydraulic control of machine tools", Pergamon press
3. Gupta A K and Arora S K, "Industrial Automation and Robotics", Third Edition, Laxmi Publication Ltd.
4. Groover M P, "Industrial Robotics Technology, programming and applications", McGraw-Hill
5. T. K. Kundra, P. N. Rao and N. K.Tewari, Numerical control and computer aided manufacturing., Tata McGraw Hill Publishing company Ltd.
6. P. Radhakrishnan, Computer numerical control machines, New Central Book Agency
7. Yoram Koren and Joseph Ben-Uri, Numerical Control of Machine Tools, Khanna Publishers, Delhi.
8. P M Agrawal and V J Patel, CNC Fundamentals and programming, Charotar Pub.
9. Software: Creo, Mastercam, CAMLAB, LABVIEW

MEUA551 : SEMINAR

M TECH 2nd SEM ADVANCED MANUFACTURING TECHNOLOGY

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	1
Marks	-	50	50	

A Outline of the Course:

- Seminar should be based on any latest topics related to the area of mechanical engineering and/or inter-disciplinary.
- It should involve exhaustive literature survey that should include books, periodicals, journals and various internet resources.
- Student should analyse the contributions in different resources and represent it in the form of seminar report in prescribed format.
- The idea behind the seminar system is to familiarize students more extensively with the methodology of their chosen subject and also to allow them to interact with examples of the practical problems that always occur during research work.

Total hours (Practical): 30

Total hours: 30

B Instructional Methods and Pedagogy:

- Students will select related topic based on subjects they learnt and other literatures like books, periodicals, journals and various internet resources.
- Students would be taught to make presentations and report based on it.
- Faculty can suggest modifications to improve technical subject matter and presentation skills.

- Each student has to prepare a write-up of about 25 pages. The report typed on A4 sized sheets and bound in the necessary format should be submitted after approved by the guide and endorsement of the Head of Department.
- The student has to deliver a seminar talk in front of the teachers of the department and his classmates. The teacher based on the quality of work and preparation and understanding of the candidate shall do an assessment of the seminar.
- At the start of course, the course delivery pattern and prerequisites will be discussed.
- Research / technical papers in relevant areas must be covered.

C Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Identify the latest trends in engineering by critical literature review and establish scope of work.

CO2: Assimilate literature on technical articles of specified topic and develop comprehension.

CO3: Design, develop and deliver presentation on specified technical topic.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	1	-	1	1
CO2	1	-	-	1
CO3	2	3	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D Recommended Study Material:

1. IEEE : www.ieeexplore.ieee.org
2. Elsevier : www.sciencedirect.com
3. Indian Jr. of Engg. and Material Sc. (www.niscair.res.in)
4. Sadhna (Engineering Science) (http://www.ias.ac.in/sadhana/)
5. Mechanical Engg. (IE)
6. Metallurgical and Materials Engg.(IE)
7. Production Engg. (IE)
8. Programming Languages: C, C++, MATLAB
9. Software’s: AutoCAD, Pro/Engineer, ANSYS, Master CAM, Hyperworks

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUE556: SURFACE ENGINEERING (Programme Elective- III)
M TECH 2nd SEM ADVANCED MANUFACTURING TECHNOLOGY

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Surface engineering	07
2.	Substrate surfaces	06
3.	Surface Preparation	07
4.	Surface deposition techniques	15
5.	Characterization of coatings and surfaces	10

Total hours (Theory): 45

Total hours (Practical): 30

Total hours: 75

B Detailed Syllabus:

1 Surface engineering — 07 Hours — 16%

1.1 Introduction to surface engineering and its terminology, classification of surface engineering processes, microstructure and properties.

1.2 Unique features of deposited materials and gaps in understanding, current applications of surface engineering, frontier areas for applications of surface engineering, selection criteria for surface engineering processes.

2 Substrate surfaces — 06 Hours — 13%

2.1 Introduction to substrate surfaces, materials for substrates.

2.2 Atomic structure and atom-particle interactions, excitation and atomic transitions, chemical bonding, probing and detected species.

2.3 Characterization of surfaces and near-surface regions, bulk properties of surfaces.

3 Surface preparation 07 Hours 16%

3.1 Introduction to surface preparation, external cleaning methods such as gross cleaning, specific cleaning and by applications of fluids.

3.2 Rinsing, drying, outgassing and outdiffusion.

3.3 Evaluating and monitoring of cleaning by various cleaning tests.

3.4 In situ cleaning, recontamination of surfaces in the ambient environment and in deposition systems.

4 Surface deposition techniques 15 Hours 33%

4.1 Introduction to Physical Vapor Deposition (PVD) processes, various types of PVD processes.

4.2 Introduction to sputtering, types of sputtering processes such as DC sputtering, Rf sputtering, magnetron sputtering, reactive magnetron sputtering, Evaporation process and apparatus, evaporation sources, thermal evaporation, electron beam evaporation.

4.3 Molecular Beam Epitaxy (MBE), ion plating, Pulsed Laser Deposition (PLD), Atomic Layer Deposition (ALD).

4.4 Introduction to Chemical Vapor Deposition (CVD) processes, important reaction zones in CVD, Classifications of CVD reactions, components of CVD systems, Types of CVD processes.

5 Characterization of coatings and surfaces 10 Hours 22%

5.1 Introduction to characterization of coatings, X-ray diffraction (XRD), Scanning Electron Microscopy (SEM), Atomic Force Microscopy (AFM), Transmission Electron Microscopy (TEM).

C Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.

- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.
- Industrial visits will be organized for students to explore industrial facilities. Students are required to prepare a report on industrial visit and submit as a part of the assignment.

D Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Modify the surface of materials by applying the coating on substrate materials.

CO2: Evaluate different deposition techniques.

CO3: Investigate the effect of various deposition parameters.

CO4: Enhance the performance of different mechanical system by applied coatings.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	-	-	1
CO2	2	1	-	2
CO3	2	-	-	2
CO4	2	1	1	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E Recommended Study Material:

Text Books/ Reference Book:

1. P.M. Martin, "Handbook of Deposition Technologies for Films and Coatings: Science, Applications and Technology", Elsevier USA (2010).
2. M. Ohring, "The Materials Science of Thin Films", Academic Press Inc, San Diego, (1992).
3. D.M. Mattox, “Handbook of Physical Vapor Deposition (PVD) Processing, Film Formation, Adhesion, Surface Preparation and Contamination Control”, Noyes Publications U.S.A., (1998).
4. H.Y. Erbil, “Surface Chemistry of Solid and Liquid Interfaces”, Blackwell Publishing Ltd, UK, (2006).
5. Satyanarayana, V.N.T. Kuchibhatla, A.S.Karakoti, D. Bera, S. Seal, “One dimensional nanostructured materials”, Progress in Materials Science 52 699–913 (2007).

6. S.C. Tjong, H. Chen, “Nanocrystalline materials and coatings”, Materials Science and Engineering R 45 1–88 (2004).
7. B. Duncan, R. Mera, D. Leatherdale, M. Taylor, R. Musgrove, "Techniques for characterising the wetting, coating and spreading of adhesives on surfaces", NPL Report DEPC MPR 020 1-42 (2005
8. A.I. Gusev, A.A. Rempel, “Nanocrystalline Materials”, Cambridge International Science Publishing UK, (2004).
9. G. Cao, "Nanostructures & Nanomaterials: Synthesis, Properties & Applications", Imperial College Press London, (2004).
10. B. Bhushan, “Springer Handbook of Nanotechnology”, Springer Berlin Heidelberg New York USA, (2004).

Reading Materials, web materials with full citations:

1. http://nptel.iitm.ac.in/courses/Webcoursecontents/IIT%20Kharagpur/Manuf%20Proc%20II/New_index1.html
2. http://www.sciencedirect.com/science/journal
3. http://www.ieindia.org/publish/pr/pr.htm
4. http://www.ieindia.info/public.asp/me/me.htm
5. http://www.ias.ac.in/sadhana

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEME551 : COMPUTATIONAL FLUID DYNAMICS
(Programme Elective- III)
M TECH 2nd SEM ADVANCED MANUFACTURING TECHNOLOGY

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction	02
2.	Basic Equations of Fluid Dynamics	10
3.	Mathematical Behavior of Partial Differential Equations	10
4.	Numerical Techniques for Discretization	15
5.	Numerical Methods	08

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1. Introduction **02 Hours** **10%**

1.1 Introduction of CFD. Applications of CFD.

1.2 Introduction of Software's for CFD.

2 Basic Equations of fluid dynamics and heat transfer **10 Hours** **20%**

2.1 Differential analysis and control volume analysis of mass, momentum and energy conservation, conduction and convection heat transfer.

2.2 Non-dimensional form of governing equations, non-dimensional parameters and their importance.

2.3	Generalized form of the governing equations, Boundary layer equations, Boundary conditions.		
3	**Mathematical Behavior of Partial Differential Equations**	**10 Hours**	**25%**
3.1	Introduction, Classification of quasi-linear partial differential equations.		
3.2	Impact of different classes of Partial differential equations (Hyperbolic, parabolic, elliptic equations) on CFD.		
4	**Numerical Techniques for Discretization**	**15 Hours**	**30%**
4.1	Introduction to Finite Differences, Taylor's expansion series, approximation techniques for first order derivative, second order derivative and mixed derivative for uniform and non-uniform grid, using forward difference scheme, backward difference scheme and central difference scheme, Steady state one-dimensional conduction equation Discretization, Steady state one dimensional conduction and convection equation analytical solution and numerical solution using forward difference, backward difference and the central difference schemes, Discretization of time dependent parameter using explicit scheme, implicit scheme, Discretization of governing equations, approximations in non-uniform grids, boundary conditions implementation, errors.		
4.2	Introduction to Finite Volume Method: Integral approach, Discretization and higher order schemes, Application to Complex Geometry.		
4.3	Introduction to Finite Element Method: Basics of finite element method, stiffness matrix, isoperimetric elements, formulation of finite elements for flow and hear transfer problems.		
5	**Numerical Methods**	**08 Hours**	**15%**
5.1	Direct Method Gauss elimination.		
5.2	Iterative Method Jacobi method, Gauss – Siedel method, Successive Over Relaxation, Convergence, merits of iterative methods over the direct methods.		

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.

- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Describe various governing equations applicable for viscous fluid flows.

CO2: Describe the mathematical behavior of partial differential equations.

CO3: Apply Finite Difference Method to one dimensional and two-dimensional Engineering problems.

CO4: Formulate and solve fluid flows and Heat transfer problems by Finite Element Method and Finite Volume Method.

CO5: Simulate the Fluid flow and heat transfer problem using analysis software.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	1	-	-	-
CO2	1	-	-	1
CO3	2	1	-	3
CO4	2	1	-	3
CO5	3	2	1	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Reference Books:

1. Patankar, “Numerical heat transfer and Fluid Flow”, Mc.GrawHill. 2002.
2. Carnahan B, “Applied numerical method” John Wiley and Sons-2001.
3. Murlidhar K and Sunderrajan T, “Computational Fluid Mechanics and Heat Transfer”, Narosa Publishing House.
4. John D Anderson Jr., “Computational Fluid Dynamics: The Basics with Applications”, McGraw-Hill, 1995.
5. Veersteeg and Malalasekara, “CFD: The Finite Volume Method by”, Prentice Hall, 1996.

6. Anderson D A, Tannehil J C and Pletcher R H, “Computational fluid mechanics and heat transfer”, Hemisphere publishing corporation,. Newyork, U. S.A2004.
7. Dante A W, “Introduction to Computational Fluid Dynamics”, Cambrige Uni. Press, 2005.
8. Reddy J N and Gartling D K, “The Finite Element Method in Heat Transfer and Fluid Dynamics”, CRC Press, 2000.
9. White F M, “Viscous Fluid Flow”, Mc Graw Hill, 1991.
10. Schlichting H, Gersten K, “Boundary-Layer Theory”, 8th edition, 2004.
11. Anderson, J D, “Modern Compressible Flow: With Historical Perspective”, McGrawHill, 2002.
12. Peric and Ferziger, “Computational Methods for Fluid Dynamics”, Springer Publication.
13. Goshdastidar, “Computer Simulation of Flow and Heat Transfer”, Tata-McGraw Hill.
14. Chandrupatla and Belegundu, “Introduction to Finite Element Methods”, Prentice Hall of India.

Web Materials:

1. http://www.nptel.iitm.ac.in/

Other Materials:

1. Programming Languages and Software’s: C, C++, MATLAB, ANSYS, ZN Tutor – CFD.
2. www.sciencedirect.com:
   International Journal of Heat and Fluid Flow.
   International Journal of Thermal Sciences.
   Experimental Thermal and Fluid Science.
3. SADHNA (Engineering Science): http://www.ias.ac.in/sadhana/
4. IEEE: www.ieeexplore.ieee.org

MEUE557: ADVANCED ENGINEERING OPTIMIZATION TECHNIQUES (Programme Elective- IV)

M TECH 2nd SEM ADVANCED MANUFACTURING TECHNOLOGY

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Fundamentals of Optimization	04
2.	Single-variable Optimization Techniques	09
3.	Multi-variable Optimization Techniques	11
4.	Constrained Optimization Techniques	10
5.	Non Traditional Optimization Techniques	11

Total hours (Theory): 45

Total hours (Practical): 30

Total hours: 75

B Detailed Syllabus:

1 Fundamentals of Optimization — 04 Hours — 08%

1.1 Introduction, statement of an optimization problem, design vector, design constraints

1.2 Single-variable and multi-variable optimization

1.3 Requirements for the application of optimization methods

1.4 Applications of optimization in engineering

2 Single-variable optimization techniques — 09 Hours — 20%

2.1 Unrestricted search, exhaustive search, dichotomous search, interval-halving method

2.2 Fibonacci method, golden-section method, quadratic interpolation method, powell’s method

2.3 Newton method, quasi-newton method, secant method

3	**Multi-variable optimization techniques**	**11 Hours**	**25%**
3.1	Simplex search method, random-walk method, univariate method, pattern search method		
3.2	Conjugate direction method, steepest descent method		
4	**Constrained optimization techniques**	**10 Hours**	**22%**
4.1	Equality and inequality constrained problems, lagrange multipliers, kuhn-tucker conditions		
4.2	Penalty concept, interior penalty function method, exterior penalty function method		
5	**Non traditional optimization techniques**	**11 Hours**	**25%**
5.1	Genetic algorithm, simulated annealing.		

C Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Apply analytical optimization techniques to solve engineering problems

CO2: Apply numerical optimization techniques to solve engineering problems

CO3: Formulate the problem at hand

CO4: Develop the computer program of the single variable optimization techniques

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	2	2	1
CO2	2	-	2	2
CO3	3	1	-	2
CO4	1	2	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E Recommended Study Material:

Text Books/ Reference Book:

1. Singeresu S. Rao, "Engineering Optimization - Theory and Practice" New Age Intl. Ltd., Publishers, 2010.
2. Kalyanamoy Deb, "Optimization for Engineering design algo rithms and Examples", Prentice Hall of India, 1995.
3. Reklaitis G.V., Ravindram A., Ragsdell K.M., "Engineering Optimization - Methods and Application", Wiley, 1993.
4. Goldberg, D.E., "Genetic algorithms in search, optimization and machine", Barnen, Addison-Wesley, New York, 1989.
5. R.L. Fox, “Optimization Methods for Engineering Design”, Addison Wesley
6. E J Haug and J S Arora, “Applied Optimal Design”, Wiley International
7. Unwubolu Godfrey C. and Babu B.V., “New Optimization Techniques in Engineering”, Springer, 2004.
8. Dennis J Jr and Schnabel R, “Numerical Methods for Unconstrained Optimization and Nonlinear Equations”, Society for Industrial and Applied Mathematics. 1996
9. Goldratt, E. M. and Cox, J., “The Goal: A Process of Ongoing Improvement”, 3rd Edition, North River Press. 2004
10. Dettmer H. William, “Goldratt's Theory of Constraints: A Systems Approach to Continuous Improvement”, American Society for Quality. 1997

Reading Materials, web materials with full citations:

1. http://nptel.iitm.ac.in/
2. Programming Languages and Software’s: Turbo C, MATLAB
3. Science Direct Journal (http://www.sciencedirect.com)
4. Mechanical Engg. (Inst. of Engineers) http://www.ieindia.org/publish/mc/mc.htm

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING

MEUE558: ARTIFICIAL INTELLIGENCE (PROGRAMME ELECTIVE IV)

(Programme Elective- IV)

M TECH 2nd SEM ADVANCED MANUFACTURING TECHNOLOGY

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to AI	04
2.	Heuristic Search Techniques	04
3.	Knowledge Representation and Logic	06
4.	Reasoning (Statistical and symbolic)	06
5.	Natural Language Processing	02
6.	Neural Networks	06
7.	Expert System	02
8.	PROLOG	04
9.	Introduction to Soft Computing	05
10.	Applications of AI in Design and Manufacturing	06

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1.	**Introduction to AI**	**04 hours**	**8%**
1.1	The AI Problems		
1.2	The Level of the Model		
1.3	Issues in the design of Search Programs		
2	**Heuristic Search Techniques**	**04 Hours**	**10%**
2.1	Generate – And- Test, Hill Climbing, Best- First Search		
2.2	Problem reduction, Constraint Satisfaction, Means-Ends Analysis		
3	**Knowledge Representation Issues**	**06 Hours**	**12%**
3.1	Knowledge Representation Issues		
3.2	Using Predicate Logic and Representing Knowledge Using Rules		
4	**Reasoning (Statistical and Symbolic)**	**06 Hours**	**12%**
4.1	Symbolic Reasoning under Uncertainty		
4.2	Statistical Reasoning and Weak Slot-And-Filler Structure		
5	**Neural Language Processing**	**02 Hours**	**05%**
5.1	Introduction, Syntactic processing, Semantic analysis, Discourse and Pragmatic processing		
6	**Neural Networks**	**06 Hours**	**15%**
6.1	Fundamentals of NN, Back propogation networks, Associative memory, Hopfield network and representations		
6.2	Learning in Neural Networks and Applications		
6.3	Connectionist AI and Symbolic AI		
7	**Expert System**	**02 Hours**	**05%**
7.1	Introduction to Expert System, Explanation Facilities, ES Development process, Knowledge acquisition		
8	**PROLOG**	**04 Hours**	**08%**
8.1	Introduction to PROLOG, Basic list manipulation function, predicates and conditional input, output and local variables		
8.2	Iteration and recursion, property list and arrays, LISP and other AI programming languages		
9	**Introduction to Soft Computing**	**05 Hours**	**10%**
9.1	Introduction to Genetic Algorithms		

9.2 Fuzzy Logic, Rough Set approach

9.3 Hybridization, Applications

10 Applications of AI in Design and Manufacturing 06 Hours 15%

10.1 Applications of AI techniques in Design and Optimization of Mechanical Components (Including Materials' selection, Dimensions and Analysis)

10.2 AI in Traditional and Non Traditional Machining Processes, Fabrications, Forming processes like Welding, Rolling, Casting, Forging etc. Process Planning and Quality Control

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.

In the lectures and laboratory discipline and behaviour will be observed strictly.

D. Course Outcomes (COs):

At the end of the course, the student will able to:

CO1: Know how machines act and thinks.

CO2: Learn the Theorem proving and Game Playing along with Reasoning, learning, planning

CO3: Understand the concepts of Neural Network-Parallel Processing, Machine learning and the behaviour of machines

CO4: Explore AI Application Areas and AI Tools.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	3	1	1	3
CO2	-	2	1	-
CO3	3	2	1	3
CO4	3	1	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

E. Recommended Study Material:

Reference Books:

1. Rich E and Knight K, "Artificial Intelligence", (2^{nd} Edition) Tata McGraw-Hill
2. Rajasekaran S and Vijayalakshmi Pai G A, "Neural Network, Fuzzy Logic and Genetic Algorithms synthesis and Applications", PHI Publication
3. Patterson D W, "Artificial Intelligence And Expert Systems "
4. Rajeshkaran S, "Hybrid Intelligent Systems", PHI
5. Townsend Carl, "Introduction to Prolog Programming"
6. Bratko Ivan, "PROLOG Programming For Artificial Intelligence", (Addison-Wesley)
7. Klocksin and Mellish, "Programming with PROLOG".
8. D. W. Rolston, "Artificial Intelligence and Expert System, Development", Mcgraw-Hill International Edition.
9. Jang J S R, Sun C T, Mizutani, "Neuro-Fuzzy and Soft Computing", Pearson Education Edition, 1997, Reprint 2007
10. Akerker and Sajja, "Knowledge-Based Systems, Jones and Bartlett", MA, USA, 2009
11. Ian Cloete and Jacek Zurada, "Knowledge Based Neuro-Computing", University Press, Massachusetts Institute of Technology, USA, 2002
12. Oscar Cordon, Francisco Herrera, Frank Hoffmann, Luis Magdalena, "Genetic Fuzzy Systems", Word Scientific Publishing Ltd. , 2001
13. Partihar D K, "Soft Computing", Narosa Publication, 2008
14. Sivanandam S N and Deepa S N, "Principles of Soft Computing", Wiley India Edition, 2007
15. Rushell and Norvig, "Modern Approach to Artificial Intelligence", Prentice Hall of India Ltd., 2006

Web Materials:

1. http://www.aaai.org/aitopics/html/overview.html
2. http://www.devarticles.com/c/a/Java/AIBased-Problem-Solving/
3. http://www.cse.unsw.edu.au/~billw/aidict.html#backwardchaining
4. http://www.cogsci.ucsd.edu/~batali/108b/lectures/heuristic.html
5. http://www.jfsowa.com/pubs/semnet.htm
6. http://www.cs.sunysb.edu/~warren/xsbbook/node3.html
7. http://www.doc.ic.ac.uk/~nd/surprise_96/journal/vol4/cs11/report.html
8. http://neurointelligence.com
9. www.soft-computing.de

Other Materials:

1. Programming Languages: PROLOG [Introduction to Turbo prolog by Townsend]
2. Software's: Alyuda Neuro Intelligence, Neuro Solutions 5.1, Easy NN Plus, Weka
3. www.ieeexplore.ieee.org
4. Computer Engineering (IE)
5. Jr. of Computing and Information science engineering
6. Jr. of Artificial Intelligence Research (AI Access Foundation)
7. Jr. of Material Processing Technology (Sciencedirect)
8. Jr. of Scientific and Industrial Research (NISCAIR, New Delhi)

MEME552: Mechanics of Fiber Reinforced Polymer Composite Structures

Description:

This course MEME552 – Mechanics of Fiber Reinforced Polymer Composite Structures is offered from SWAYAM.

Credit and Week:

Teaching Scheme	Week	Marks	Credit
	12	100	4

About this course:

This is introductory course on Mechanics of Fiber Reinforced Composite Structures. One course is basically aimed at introducing the students of mechanical/civil engineering streams to the basics of design and analysis of structural components made of FRP composites. The contents of the course is so designed that it requires the first course on Strength of Materials/ Solid Mechanics as a prerequisite which is anyway a core course for mechanical/civil undergraduates. It introduces the students first to the basic mechanics (stress strain and load deformation relations) of fiber composites, possible failure modes and corresponding failure theories proposed. Next, the course introduces the design and analysis using those concepts along with the design of some components made of such materials. At the end a few topics of slightly advanced nature (for UG students) are kept for brief introduction only.

Prerequisites:

No specific pre-requisite. Fundamental knowledge of Strength of Materials / Solid Mechanics.

INDUSTRY SUPPORT:

DRDO, ISRO, NAL

Course layout:

Week 1: Introduction to FRP Composites
Week 2: Review of Elasticity
Week 3: Macromechanics of Lamina - I
Week 4: Macromechanics of Lamina - II
Week 5: Micromechanics of Lamina - I
Week 6: Micromechanics of Lamina - II
Week 7: Elastic Behaviour of Laminates - I
Week 8: Elastic Behaviour of Laminates - II
Week 9: Failure Analysis of Laminates
Week 10: Design Examples

Week 11: Interlaminar Stresses
Week 12: Transverse Deflection, Buckling and Free vibration of Laminated Plates

Books and references:

1. Robert M Jones, Mechanics Of Composite Materials, 2nd Edition, CRC Press
2. Autar Kaw, Mechanics of Composite Materials, 2nd Edition, Taylor and Francis
3. I M Daniel and O Ishai, Engineering Mechanics of Composite Materials, 2nd Editiory Oxford University Press

Course certificate:

- The course is free to enroll and learn from. But if you want a certificate, you have to register and write the proctored exam conducted by us in person at any of the designated exam centres.
- The exam is optional for a fee of Rs 1000/- (Rupees one thousand only).
- Date and Time of Exams: Announcements will be made once the course launch.
- Registration url: Announcements will be made when the registration form is open for registrations.
- The online registration form has to be filled and the certification exam fee needs to be paid.
- More details will be made available when the exam registration form is published. If there are any changes, it will be mentioned then.

Please check the form for more details on the cities where the exams will be held, the conditions you agree to when you fill the form etc.

CRITERIA TO GET A CERTIFICATE

Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.
Exam score = 75% of the proctored certification exam score out of 100

Final score = Average assignment score + Exam score

YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE >=10/25 AND EXAM SCORE >= 30/75. If one of the 2 criteria is not met, you will not get the certificate even if the Final score >= 40/100.

Certificate will have your name, photograph and the score in the final exam with the breakup.It will have the logos of NPTEL and IIT Guwahati. It will be e-verifiable at nptel.ac.in/noc.

Only the e-certificate will be made available. Hard copies will not be dispatched.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
HSUA501: ACADEMIC WRITING
M TECH 2nd SEM ADVANCED MANUFACTURING TECHNOLOGY

Credits and Schemes:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Academic Writing and Research Process	05
2.	Anatomy of Academic Writing	05
3.	Key Academic Skills	05
4.	Accuracy in Academic Writing	05
5.	Using and Citing Sources of Ideas	05
6.	Contemporary Practices in Academic Writing	05

Total hours (Theory): 00
Total hours (Practical): 30
Total hours: 30

B. Detailed Syllabus:

1.	**Academic Writing and Research Process**	5 Hours
	Introduction to Academic Writing, Academic Writing as a Part of Research, Types of Academic Writing, Features of Academic Writing, Importance of Good Academic Writing in various Academic Works	
2.	**Anatomy of Academic Writing**	5 Hours
	Academic Vocabulary, Simple and Complex Sentences, Organizing Paragraphs, The Writing Process, Adopting Academic Writing Style	
3.	**Key Academic Skills**	5 Hours
	Note – taking, Note – making, Paraphrasing, Summarizing	
4.	**Accuracy in Academic Writing**	5 Hours

	Lexical Range, Academic Language and Structures, Elements of Writing, Proof Reading, Editing, and Rewriting	
5.	**Using and Citing Sources of Ideas**	5 Hours
	Academic Texts and their Types, Intellectual Honesty in Academic Writing, Avoiding Plagiarism – Idea Theft, Degrees of Plagiarism, Types of Borrowing, Anatomy of Citations, Common Citation Styles	
6.	**Contemporary Practices in Academic Writing**	5 Hours
	Analytical Essays, Graph / Table / Process Interpretation and Description, Writing Reports and Abstract, Writing Research / Concept Papers	

C. Instructional Method and Pedagogy:

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like writing, group work, independent and collaborative research, etc.

D. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sr. No.	Component	Number	Marks per incidence	Total Marks
1	Paragraph Writing	1	3	03
2	Note-taking / Note-making	1	3	03
3	Paraphrasing / Summarizing	1	4	04
4	Essay Writing	1	5	05
5	Concept Paper Writing	1	10	10
5	Attendance and Class Participation			05
			Total	**30**

External Evaluation

The University Practical Examination will be for 70 marks and will test the professional communication skills and academic writing.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical /Quiz/ Project / Academic Writing	-	70	70
			Total	**70**

E. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Explain the concept and applications of academic writing

CO2: Acquire enough knowledge of academic writing style, strategy and approach.

CO3: Demonstrate error free and effective academic writing.

CO4: Demonstrate ability to work on project/report/paper writing.

CO5: Understand the Research and Research Methodology.

CO6: Effectively communicate in diverse academic and professional settings.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	1	3	1	1
CO2	-	3	-	-
CO3	-	3	-	-
CO4	-	3	-	-
CO5	3	3	-	-
CO6	-	3		

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

F. Recommended Study Material:

Essential Reading for Concepts

- Academic Writing for International Students, Routledge
- Academic Writing: A Guide for Management Students and Researchers. Monipally, M. M. & Pawar, B. S. Sage. 2010. New Delhi

Essential Reading for Activity and Teacher Resource

- ***Effective Academic Writing Level - 1,2,3,4 (Second Edition) By:*** Alice Savage, Patricia Mayer, Masoud Shafiei, Rhonda Liss, & Jason Davis; *Publisher: Oxford*

Additional Reading

- Writing Your Thesis (2nd Edition) by Paul Oliver, Sage
- Development Communication In Practice by Vilanilam V J, Sage
- Intercultural Communication by Mingsheng Li, Patel Fay, Sage
- www.owl.perdue.edu

M. Tech. (AMT) Programme

SYLLABI
(Semester – 3)

MEUP601 : PROJECT PRELIMINARIES

M TECH 3rd SEM ADVANCED MANUFACTURING TECHNOLOGY

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	-	-	4
Marks	-	-	100	

A Outline of the Course:

- Project Preliminaries include course work on a specialized Subject or a Seminar.
- The course work shall be related to the area of his/her project research work.
- The coursework may be chosen from the existing PG (M. Tech.) Programmes of the registering department or from those of other departments.
- In addition to existing courses of the M. Tech. Programme, a department may offer special courses or seminar topics to the students.
- The specific subject of course work study will be decided by the department level Post Graduate Committee on recommendation of the supervisor(s).
- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Resident students will satisfy the course requirement by attending Institute and Departmental courses, and submitting technical writings on assigned course.
- Non-resident students (full time research in R&D institutions/industries) will submit equivalent amount of technical writing and reports on seminars from their own places of work.
- Course offered at 3rd Semester will be evaluated at least once during the semester and at the end of the semester as a part of continuous evaluation.
- The course work and report is expected to show clarity of thought and expression, critical appreciation of existing literature and analytical or experiment or software based applications in the field of specialization.
- Course work can be based on topic of the Project Work. It may include literature review,

required theoretical input, study and comparison of various approaches for the proposed project work.

- A student has to produce some useful outcome by conducting experiments or simulations.
- Student can learn all aspects and functionality of specialized software in the institute.
- Student can generate one working test bench using research paper along with results verified from referred paper through simulation and by modify existing work student can produce one research paper (research studies)

B Instructional Methods and Pedagogy:

- Students will select related topic based on subjects they learnt and other literatures like books, periodicals, journals and various internet resources.
- Students can select the topic based on the research areas of available supervisor/guide.
- Each student has to prepare a write-up of about 30-40 pages in prescribed format only. The report typed on A4 sized sheets and bound should be submitted after approval by the guide and endorsement of the Head of Department.
- Performance of students at the seminar will be assessed by the department level Post Graduate Committee. The supervisor/guide based on the quality of work and preparation and understanding of the candidate shall do an assessment of the course.

C Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Explore current issues and research areas.

CO2: Learn the additional technical skill useful for the dissertation work.

CO3: Explain various research methodologies

CO4: Develop technical writing and communication skill.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	-	1	-
CO2	3	-	-	1
CO3	2	-	-	2
CO4	-	3	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D Recommended Study Material:

1. Programming Languages: C, C^{++}, MATLAB
2. Softwares: AutoCAD, Pro/Engineer, ANSYS, Hyperworks, Master CAM, Rapid XOR
3. IEEE : www.ieeexplore.ieee.org
4. www.sciencedirect.com
5. Indian Jr. of Engg. and Material Sc. (www.niscair.res.in)
6. Sadhna (Engineering Science) (http://www.ias.ac.in/sadhana/)
7. Mechanical Engg. (IE), Metallurgical and Materials Engg.(IE),
   Production Engg. (IE)

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CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUP602 : PROJECT PHASE I
M TECH 3rd SEM ADVANCED MANUFACTURING TECHNOLOGY

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	-	-	18
Marks	-	-	500	

A Outline of the Course:

- The Project shall be related to the major field of his/her PG specialization work.
- The Project should be one of the major pieces of evidence that students are familiar with or that student wants to be familiar with. It should reflect specialist subject by means of deep and sustained study.
- The project will be finalized by the department level Post Graduate Committee on recommendation of the supervisor(s).
- The project work shall be carried out by each candidate independently during the third and fourth semester under the guidance of one of the faculty members of the Department. If the project work is of inter-disciplinary nature, a co-guide shall be taken from the same or any other relevant Department.
- If a project work has to be carried out in any industry / factory / organization, outside the campus, the permission to that effect and the name of co-guide at any of these organizations shall be intimated to the Post Graduate Committee at the beginning of third semester.
- Project I includes literature review, required theoretical input, study and comparison of various approaches for the proposed dissertation work.

B Instructional Methods and Pedagogy:

- Student has to submit a project/dissertation proposal indicating the tentative title and broad outline of the proposed work and the name(s) of the supervisor(s) along-with their concurrence in writing within 30 days from the starting of the third semester.

- Utmost care should be taken in selection of research topic so that repetition of research work is avoided.
- Project - I will be evaluated at least once during the semester and at the end of the semester as a part of continuous evaluation.
- After successful completion of Project I only students are allowed to go register for Project – II.

C Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Apply concept to formulate the research problem.

CO2: Identify the research area by critical literature review and establish scope of work.

CO3: Develop study / experimental methodology.

CO4: Develop technical writing and communication skill.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	3	-	-	3
CO2	2	-	2	2
CO3	3	-	-	3
CO4	-	3	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D Recommended Study Material:

1 Programming Languages: C, C^{++}, MATLAB

2 Softwares: AutoCAD, Pro/Engineer, ANSYS, Hyperworks, Master CAM, Rapid XOR

3 IEEE : www.ieeexplore.ieee.org

4 www.sciencedirect.com

5 Indian Jr. of Engg. and Material Sc. (www.niscair.res.in)

6 Sadhna (Engineering Science) (http://www.ias.ac.in/sadhana/)

7 Mechanical Engg. (IE), Metallurgical and Materials Engg.(IE),
Production Engg. (IE)

M. Tech. (AMT) Programme

SYLLABI
(Semester – 4)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
MEUP603 : PROJECT PHASE II
M TECH 4th SEM ADVANCED MANUFACTURING TECHNOLOGY

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	-	-	18
Marks	-	-	500	

A Outline of the Course:

- Student should carry out the investigation by identifying sources of evidence, accessing those using accepted and rigorous academic methods, and analysing and interpreting the material gathered by simulation/experimentation.
- A project - II is student's own work and will need to keep up the effort, and the interest, over several months and through several stages.
- Student need to think carefully about the time necessary to carry-out and complete your project work and the relative writing up.
- The project should present an orderly and critical exposition of the existing knowledge of the subject and will embody results of original investigations demonstrating the capacity of the candidate to do independent research work.
- While writing the thesis/dissertation, the candidate will layout clearly the work done by him independently and the sources from which he has obtained other information contained in his/her Dissertation

B Instructional Methods and Pedagogy:

- Project - II will be evaluated at least once during the semester and at the end of the semester as a part of continuous evaluation.
- Before submission of Phase II project/dissertation report, it is expected from a student to publish at least one research paper in National/International conference. Further, for such

publications, Department Post Graduate Committee will identify and approve the national/international conferences.

- The dissertation shall be submitted for ‘dissertation – evaluation’ ordinarily at the end of IV Semester and ‘dissertation – open defence’ shall be held soon after the submission of the dissertation.

C Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Design and create the experiment setup to collect data.

CO2: Analyse and conclude experimental results.

CO3: Write and publish the technical research paper.

CO4: Develop technical writing and communication skill.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	3	-	2	3
CO2	3	-	-	3
CO3	-	3	1	1
CO4	-	3	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D Recommended Study Material:

1 Programming Languages: C, C^{++}, MATLAB

2 Softwares: AutoCAD, Pro/Engineer, ANSYS, Hyperworks, Master CAM, Rapid XOR

3 IEEE : www.ieeexplore.ieee.org

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7 Mechanical Engg. (IE), Metallurgical and Materials Engg.(IE), Production Engg. (IE)